

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

8104

Nombre de lignes df clean	8104
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5672
Taille dataset test	2431
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_mean, r_NH3_st_2h_std, r_NH3_st_2h_rms, r_NH3_st_2h_median, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_umidity_st_2h_mean, r_umidity_st_2h_std, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max, r_umidity_st_2h_diff_last, r_umidity_st_2h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

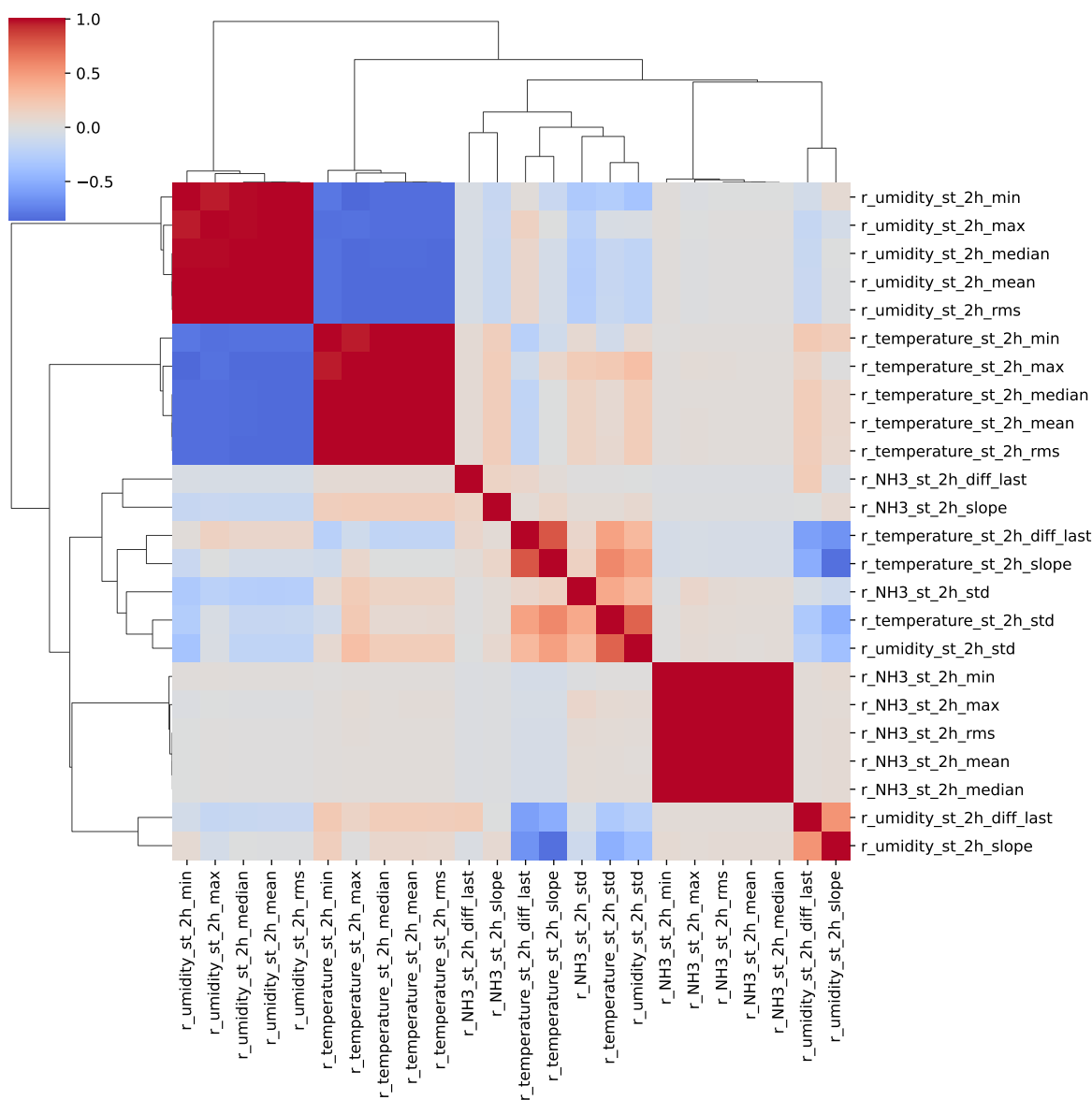
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_2h_min, r_NH3_st_2h_mean, r_NH3_st_2h_max, r_NH3_st_2h_median, r_NH3_st_2h_rms ; **cluster 7:** r_NH3_st_2h_std ; **cluster 8:** r_NH3_st_2h_diff_last ; **cluster 9:** r_NH3_st_2h_slope ; **cluster 10:** r_temperature_st_2h_mean, r_umidity_st_2h_mean, r_temperature_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max, r_temperature_st_2h_max, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_rms ; **cluster 11:** r_temperature_st_2h_std ; **cluster 12:** r_temperature_st_2h_diff_last ; **cluster 13:** r_umidity_st_2h_slope,

r_temperature_st_2h_slope ; **cluster 14:** r_umidity_st_2h_std ; **cluster 15:** r_umidity_st_2h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_tempera-

ture_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last

Len(Features_X): 15

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

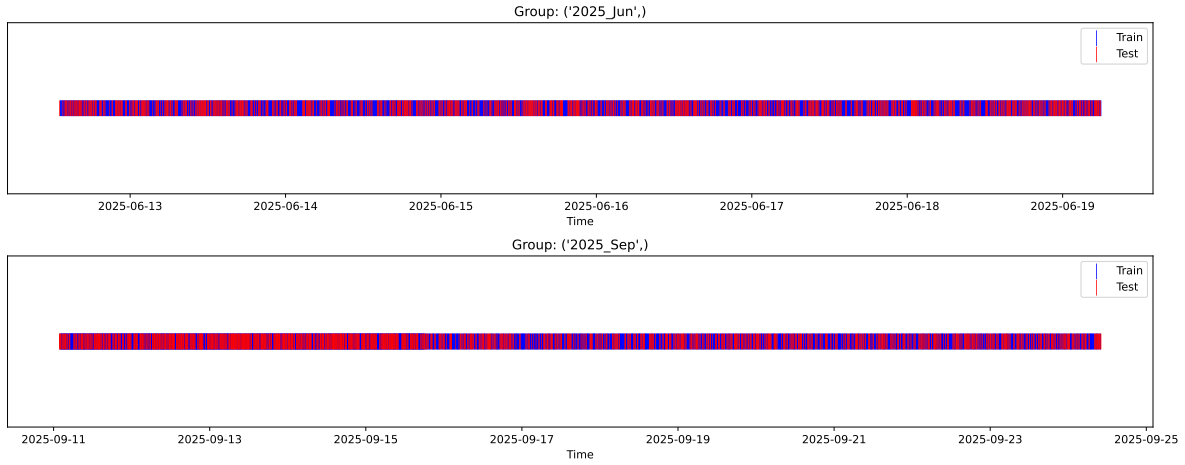
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

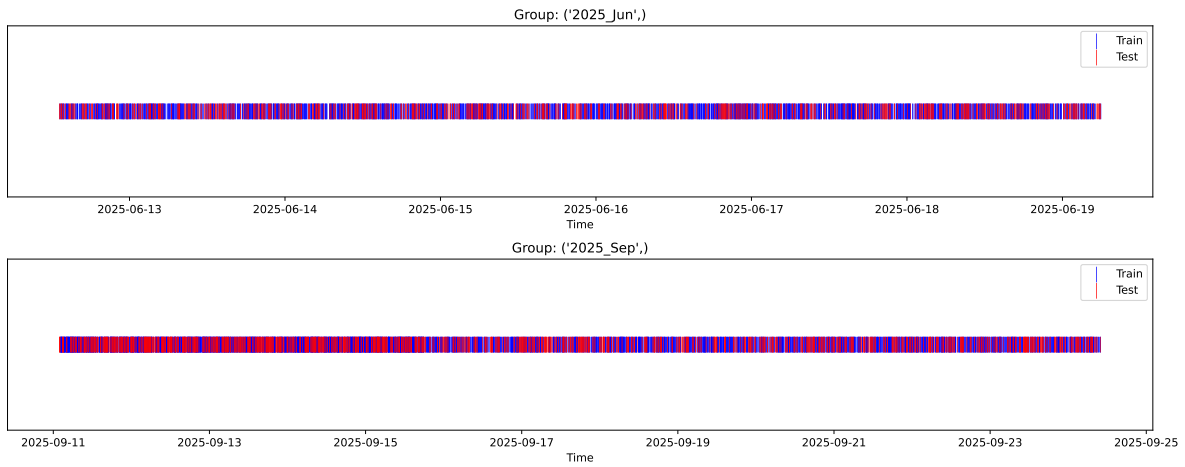
External Split: Train: 5672 samples Test: 2432 samples



3.3.2 Internal CV Folds

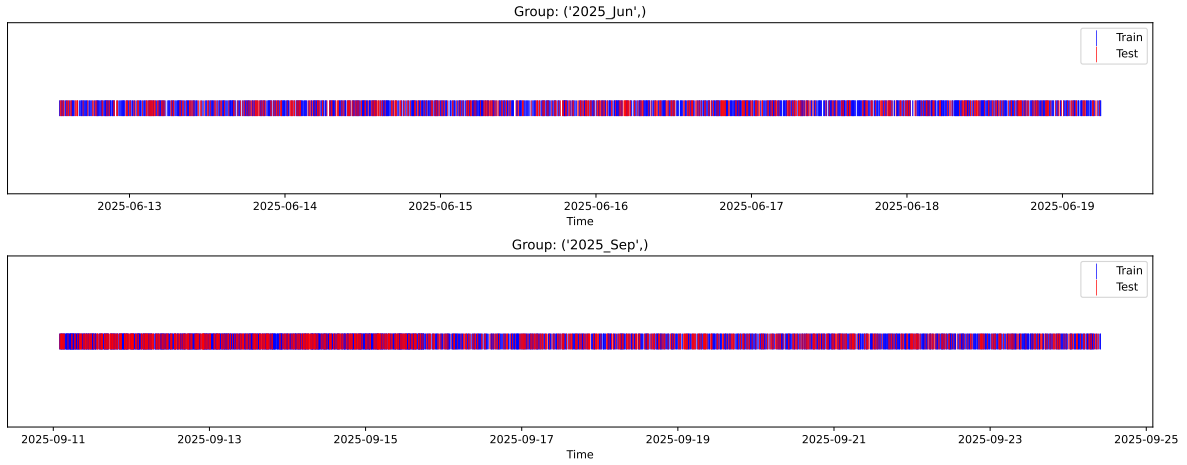
Fold 1/5

Train: 3970 samples Test: 1702 samples Excluded: 0 samples Window exclusion: 0h



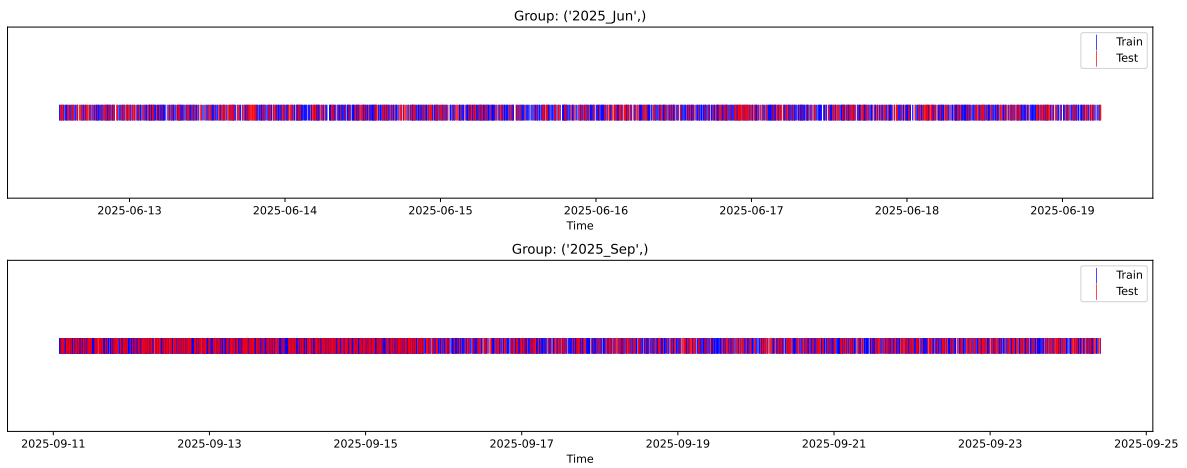
Fold 2/5

Train: 3970 samples Test: 1702 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3970 samples Test: 1702 samples Excluded: 0 samples Window exclusion: 0h



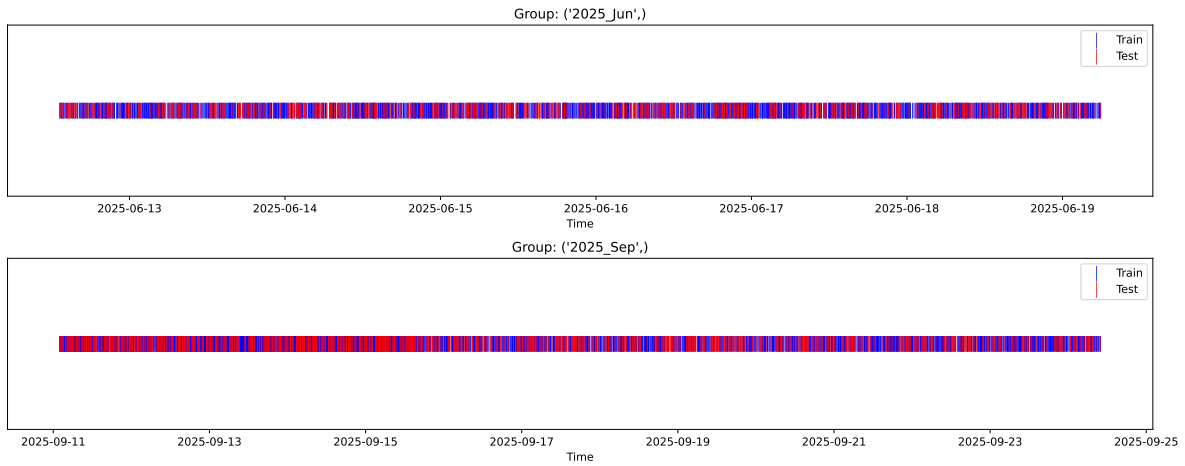
Fold 4/5

Train: 3970 samples Test: 1702 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3970 samples Test: 1702 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.344, std=0.794, min=0.973, max=13.845
 y_{test} : mean=2.367, std=0.800, min=0.959, max=8.682

Fold 0 — y_{test} : mean=2.35, std=0.80, y_{train} : mean=2.34, std=0.79

Fold 1 — y_{test} : mean=2.32, std=0.76, y_{train} : mean=2.35, std=0.81

Fold 2 — y_{test} : mean=2.34, std=0.78, y_{train} : mean=2.34, std=0.80

Fold 3 — y_test: mean=2.37, std=0.80, y_train: mean=2.33, std=0.79

Fold 4 — y_test: mean=2.35, std=0.81, y_train: mean=2.34, std=0.79

train scores CV: [0.54647849 0.5421662 0.5567357 0.5416047 0.53790437]

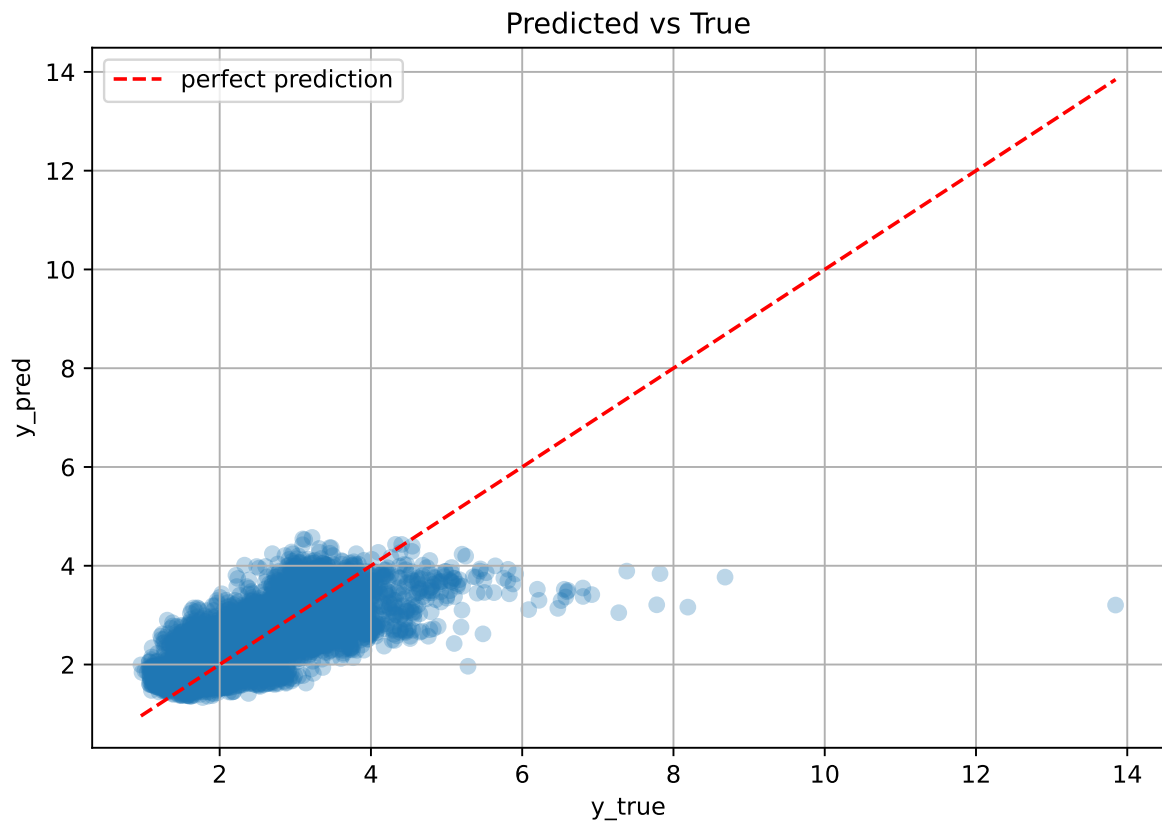
test scores CV: [0.56125467 0.57217057 0.53669479 0.57222495 0.58017534]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.545	0.5645	0.0152	0.5369	0.5374	-0.0271	0.9519	0.9654

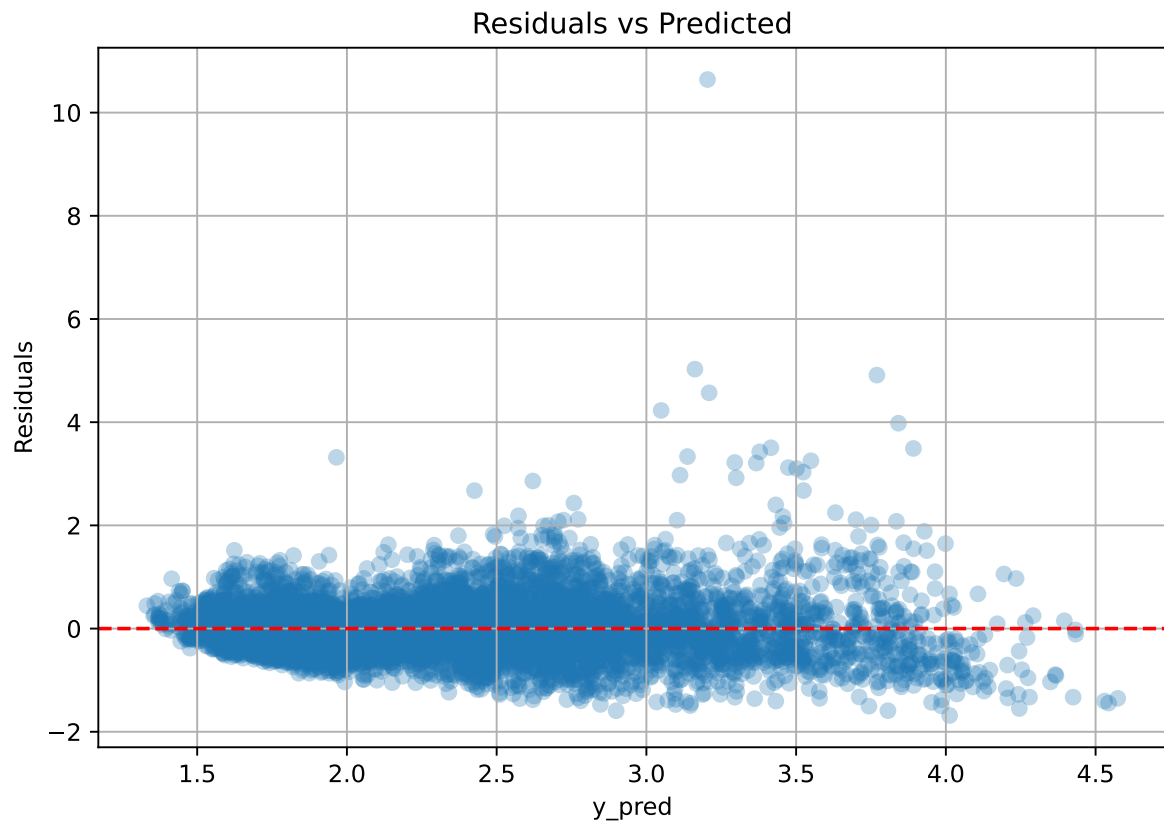
Coefficients:

coef_r_CO2 = -0.0486, coef_r_NH3 = 0.1250, coef_r_NH3_st_2h_diff_last = 0.0191, coef_r_NH3_st_2h_min = -0.1704, coef_r_NH3_st_2h_slope = 0.0062, coef_r_NH3_st_2h_std = 0.0540, coef_r_THI = 0.3675, coef_r_temperature = 0.2855, coef_r_temperature_st_2h_diff_last = -0.0286, coef_r_temperature_st_2h_mean = -0.6851, coef_r_temperature_st_2h_std = 0.1062, coef_r_umidity = -0.4213, coef_r_umidity_st_2h_diff_last = -0.0413, coef_r_umidity_st_2h_slope = -0.0528, coef_r_umidity_st_2h_std = 0.0317, intercept = 2.3440,

3.4.2 Scatter



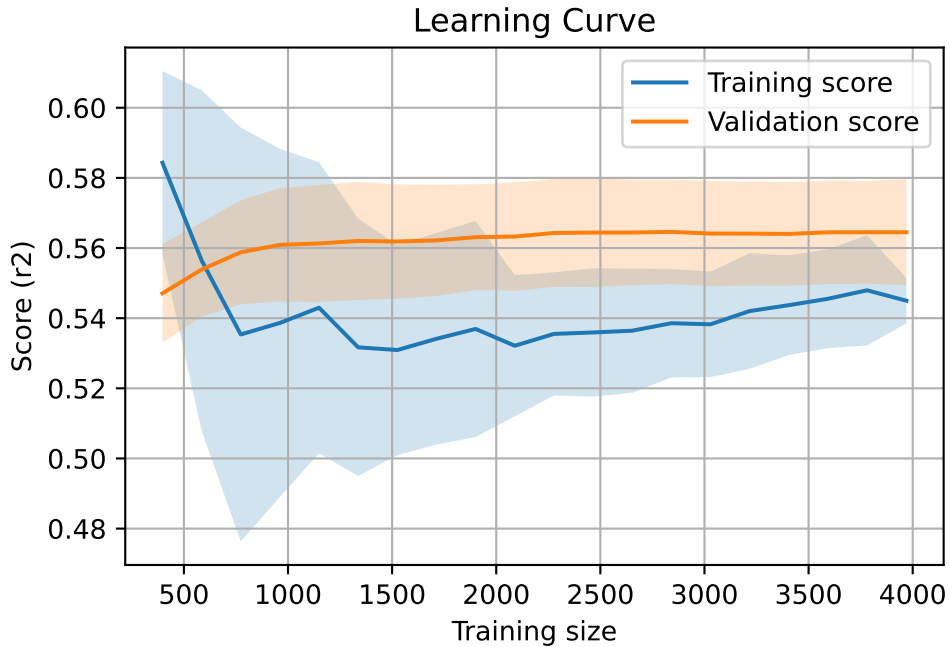
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 8104, **Len(training)** : 5673,

Len(training_real) : 3972, **Len(test_of_training)** : 1701, **Len(test)** : 2431

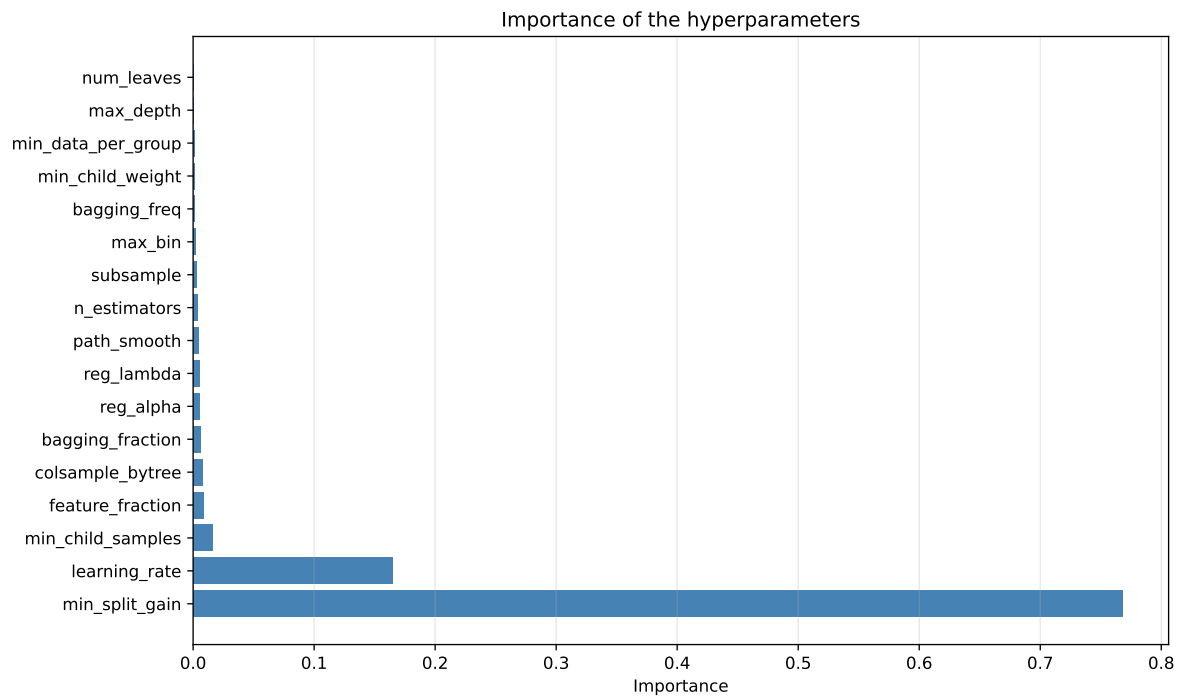
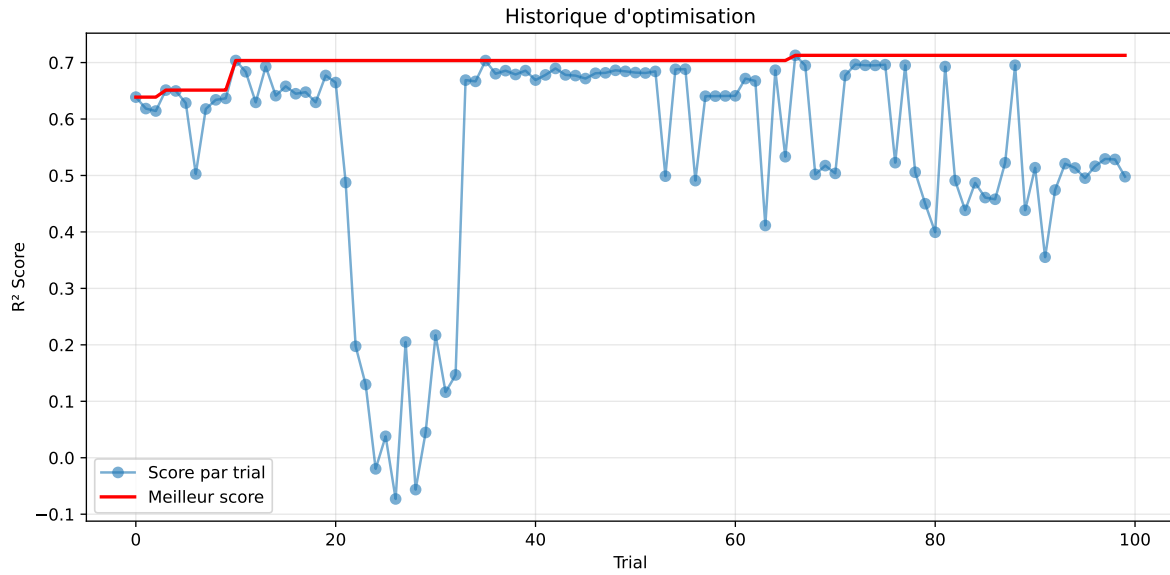


3.5 Model : LGBM

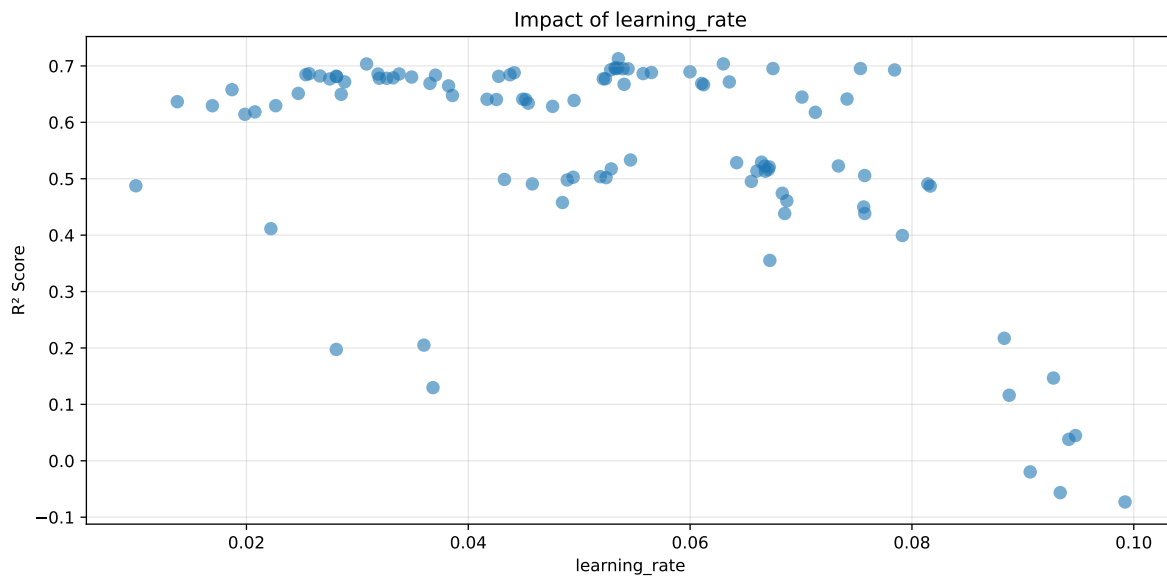
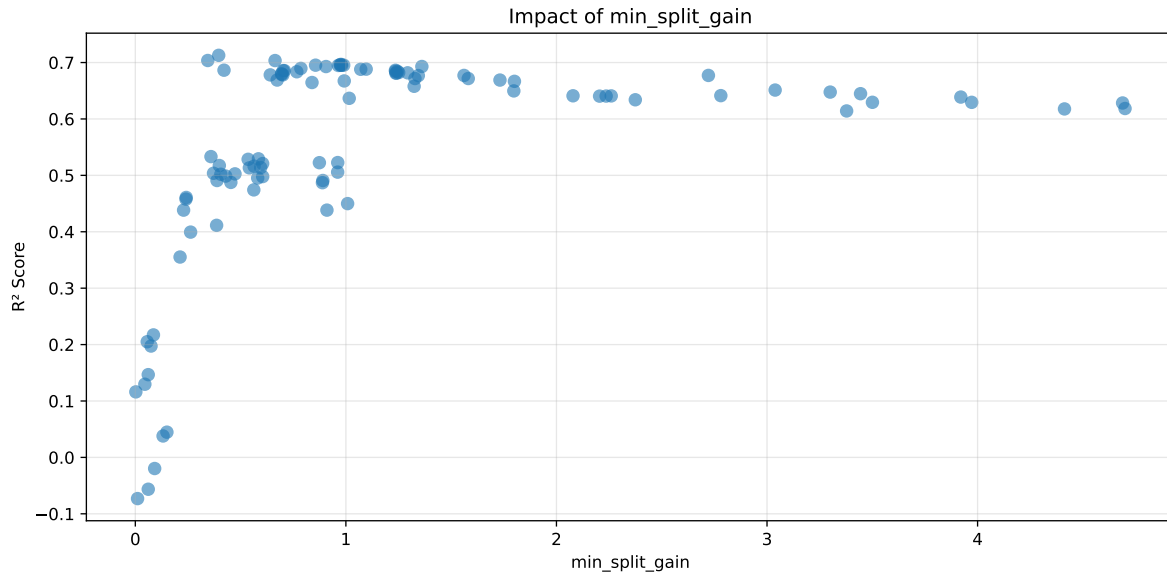
3.5.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.344, std=0.794, min=0.973, max=13.845
 y_{test} : mean=2.367, std=0.800, min=0.959, max=8.682

===== TOP Importance FEATURES =====
 feature importance r_CO2 428 r_NH3 264 r_NH3_st_2h_min 249 r_umidity 211 r_umid-
 ity_st_2h_slope 178 r_temperature_st_2h_std 150 r_temperature_st_2h_diff_last
 147 r_NH3_st_2h_slope 144 r_temperature 139 r_NH3_st_2h_std 122 r_tempera-
 ture_st_2h_mean 105 r_umidity_st_2h_std 101 r_THI 88 r_NH3_st_2h_diff_last 75
 r_umidity_st_2h_diff_last 70

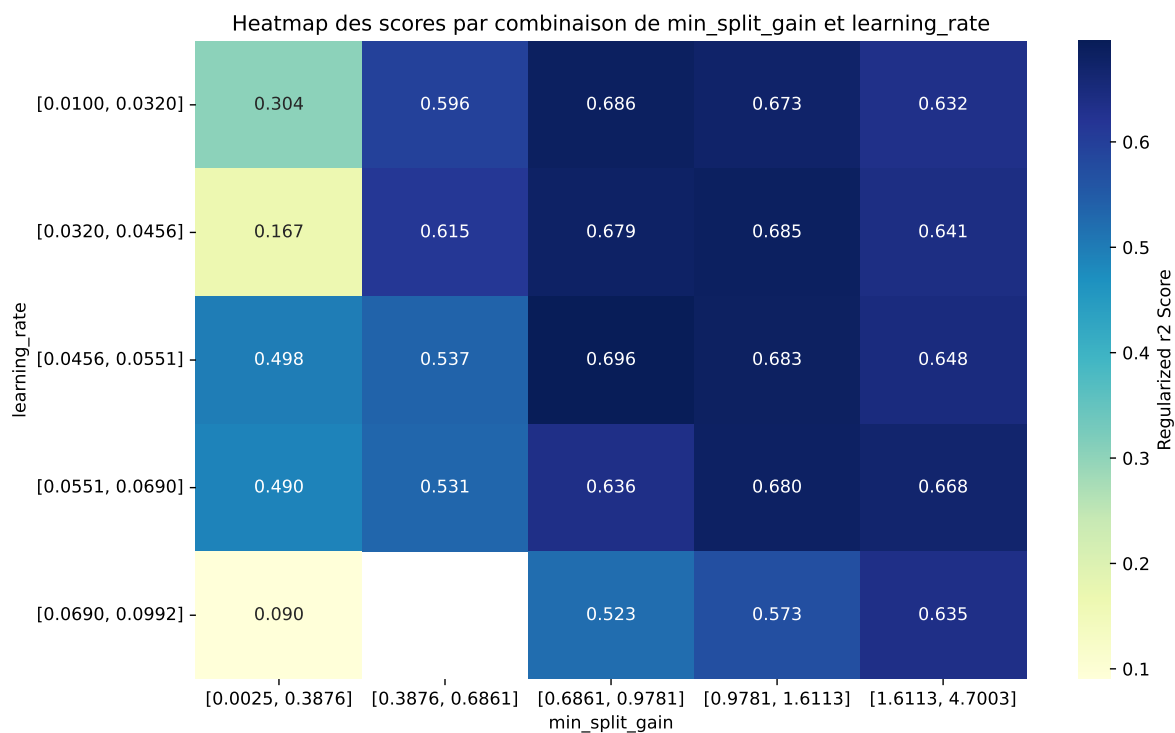


Paramètres avec >5% d'importance : ['min_split_gain', 'learning_rate']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of `observed=False` is deprecated and will change to `observed=True` in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7477	0.7128	0.6777	0.0163	0.7128	0.6938	-0.0189	0.9734	1.049
2	0.7384	0.7036	0.6698	0.0142	0.7036	0.6892	-0.0144	0.9795	1.0494
3	0.7335	0.7035	0.6727	0.0142	0.7035	0.6897	-0.0138	0.9804	1.0426
4	0.7262	0.6967	0.6632	0.016	0.6967	0.6808	-0.0159	0.9772	1.0424
5	0.7278	0.6962	0.6615	0.0155	0.6962	0.678	-0.0182	0.9739	1.0454
6	0.7237	0.6953	0.6617	0.0149	0.6953	0.6765	-0.0188	0.973	1.0408
7	0.7232	0.6953	0.6608	0.015	0.6953	0.6791	-0.0163	0.9766	1.0402
8	0.7185	0.6952	0.6611	0.016	0.6952	0.6789	-0.0163	0.9765	1.0336
9	0.72	0.695	0.6604	0.0135	0.695	0.6766	-0.0185	0.9734	1.0359
10	0.7225	0.6948	0.6602	0.0128	0.6948	0.6845	-0.0103	0.9852	1.0399
---	---	---	---	---	---	---	---	---	---
100	0.9213	0.7556	0.7322	0.0149	-0.073	0.7542	-0.0013	0.9982	1.2193

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.05353176243968611, 'reg__num_leaves': 54, 'reg__subsample': 0.730212550918677, 'reg__colsample_bytree': 0.5319719740723078, 'reg__min_child_samples': 40, 'reg__reg_alpha': 3.5011371935049636, 'reg__reg_lambda': 15.215875515142349, 'reg__min_split_gain': 0.3959882260256418, 'reg__path_smooth': 3.0442040631777068, 'reg__min_child_weight':

0.16303696144232419, 'reg__feature_fraction': 0.5111379705072139, 'reg__bagging_fraction': 0.6971228801752517, 'reg__bagging_freq': 4, 'reg__max_bin': 68, 'reg__min_data_per_group': 128}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.0629941297651093, 'reg__num_leaves': 38, 'reg__subsample': 0.6034555189443804, 'reg__colsample_bytree': 0.8629023917370102, 'reg__min_child_samples': 35, 'reg__reg_alpha': 8.157870698922272, 'reg__reg_lambda': 30.622833379074088, 'reg__min_split_gain': 0.3441183115566526, 'reg__path_smooth': 3.5988687685807803, 'reg__min_child_weight': 7.944485840866747, 'reg__feature_fraction': 0.8560308957344744, 'reg__bagging_fraction': 0.7583370241786187, 'reg__bagging_freq': 1, 'reg__max_bin': 143, 'reg__min_data_per_group': 120}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 7, 'reg__learning_rate': 0.030830395473915383, 'reg__num_leaves': 34, 'reg__subsample': 0.6047823709505342, 'reg__colsample_bytree': 0.716131624305144, 'reg__min_child_samples': 40, 'reg__reg_alpha': 1.3234082561169604, 'reg__reg_lambda': 23.754363908923168, 'reg__min_split_gain': 0.6636394045190401, 'reg__path_smooth': 3.591134515540037, 'reg__min_child_weight': 0.001124326629884936, 'reg__feature_fraction': 0.6598845516839822, 'reg__bagging_fraction': 0.512802797202913, 'reg__bagging_freq': 3, 'reg__max_bin': 120, 'reg__min_data_per_group': 148}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.05322737742023868, 'reg__num_leaves': 40, 'reg__subsample': 0.5509406961115321, 'reg__colsample_bytree': 0.5477107326890838, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.6310780655044512, 'reg__reg_lambda': 9.755603528035103, 'reg__min_split_gain': 0.9766376455883585, 'reg__path_smooth': 3.7602498714679986, 'reg__min_child_weight': 0.03552412881413848, 'reg__feature_fraction': 0.5214327110766543, 'reg__bagging_fraction': 0.7745328577650487, 'reg__bagging_freq': 5, 'reg__max_bin': 69, 'reg__min_data_per_group': 127}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.053487120151268094, 'reg__num_leaves': 39, 'reg__subsample': 0.7324539626551063, 'reg__colsample_bytree': 0.5354217256771949, 'reg__min_child_samples': 11, 'reg__reg_alpha': 2.4825754330687864, 'reg__reg_lambda': 9.633334066389674, 'reg__min_split_gain': 0.9728167943922166, 'reg__path_smooth': 4.954780685705305, 'reg__min_child_weight': 0.020055528903962266, 'reg__feature_fraction': 0.5176817992123712, 'reg__bagging_fraction': 0.7731430850327449, 'reg__bagging_freq': 5, 'reg__max_bin': 73, 'reg__min_data_per_group': 127}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.06746903341315488, 'reg__num_leaves': 38, 'reg__subsample': 0.7560139324969191, 'reg__colsample_bytree': 0.5509110718106229, 'reg__min_child_samples': 11, 'reg__reg_alpha': 2.5786586797655806, 'reg__reg_lambda': 26.421659863799896, 'reg__min_split_gain': 0.855985603377275, 'reg__path_smooth': 3.291112914498831, 'reg__min_child_weight':

0.015088697585622205, 'reg__feature_fraction': 0.5096611359243626, 'reg__bagging_fraction': 0.8066095655214239, 'reg__bagging_freq': 6, 'reg__max_bin': 79, 'reg__min_data_per_group': 144}

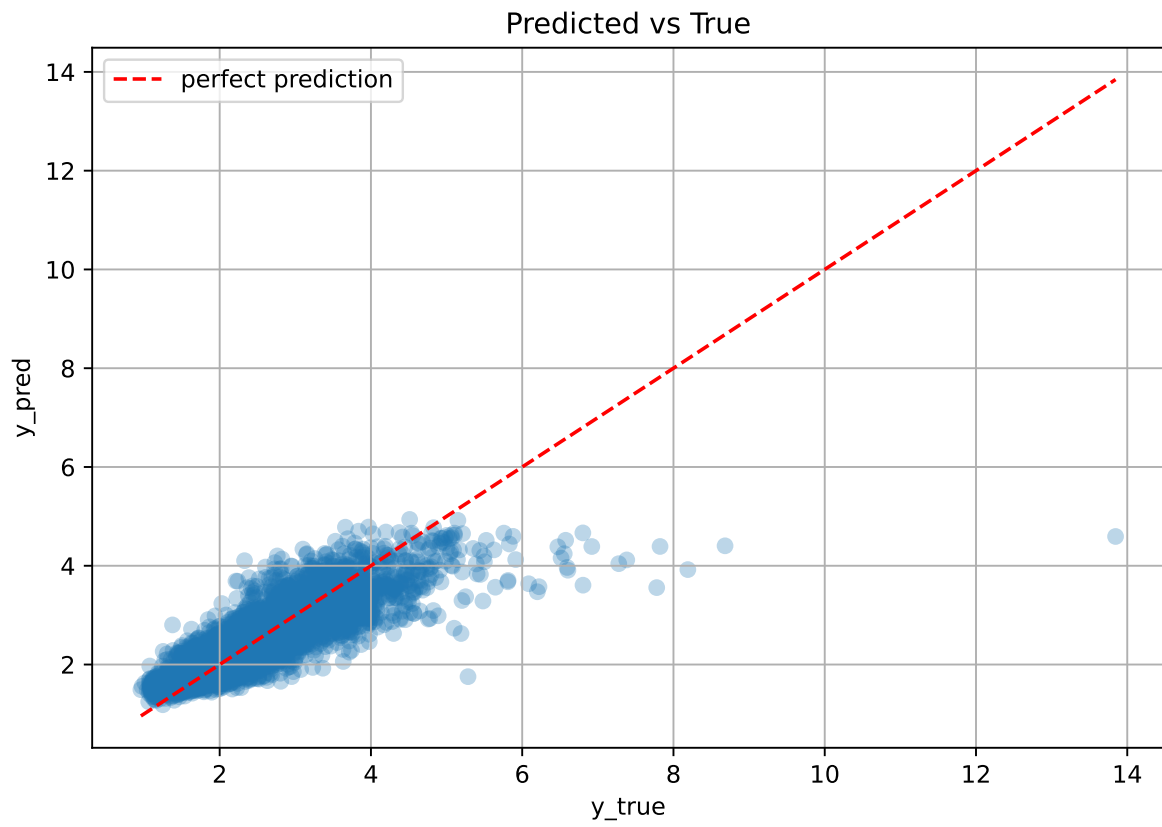
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.07535428924508801, 'reg__num_leaves': 39, 'reg__subsample': 0.7151544217958248, 'reg__colsample_bytree': 0.5547274476593079, 'reg__min_child_samples': 29, 'reg__reg_alpha': 2.4973525652617234, 'reg__reg_lambda': 26.320963135015152, 'reg__min_split_gain': 0.9895617070769606, 'reg__path_smooth': 0.24841405052330878, 'reg__min_child_weight': 0.04177487514102868, 'reg__feature_fraction': 0.5322536720468951, 'reg__bagging_fraction': 0.891870904182706, 'reg__bagging_freq': 5, 'reg__max_bin': 75, 'reg__min_data_per_group': 129}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.05330141263695253, 'reg__num_leaves': 40, 'reg__subsample': 0.5550552187477373, 'reg__colsample_bytree': 0.5433276489895853, 'reg__min_child_samples': 44, 'reg__reg_alpha': 2.5430079139833275, 'reg__reg_lambda': 10.633609944504489, 'reg__min_split_gain': 0.9802008846786461, 'reg__path_smooth': 3.7169710140090557, 'reg__min_child_weight': 0.020020604511154214, 'reg__feature_fraction': 0.5222745810169371, 'reg__bagging_fraction': 0.7733345523945224, 'reg__bagging_freq': 5, 'reg__max_bin': 69, 'reg__min_data_per_group': 50}

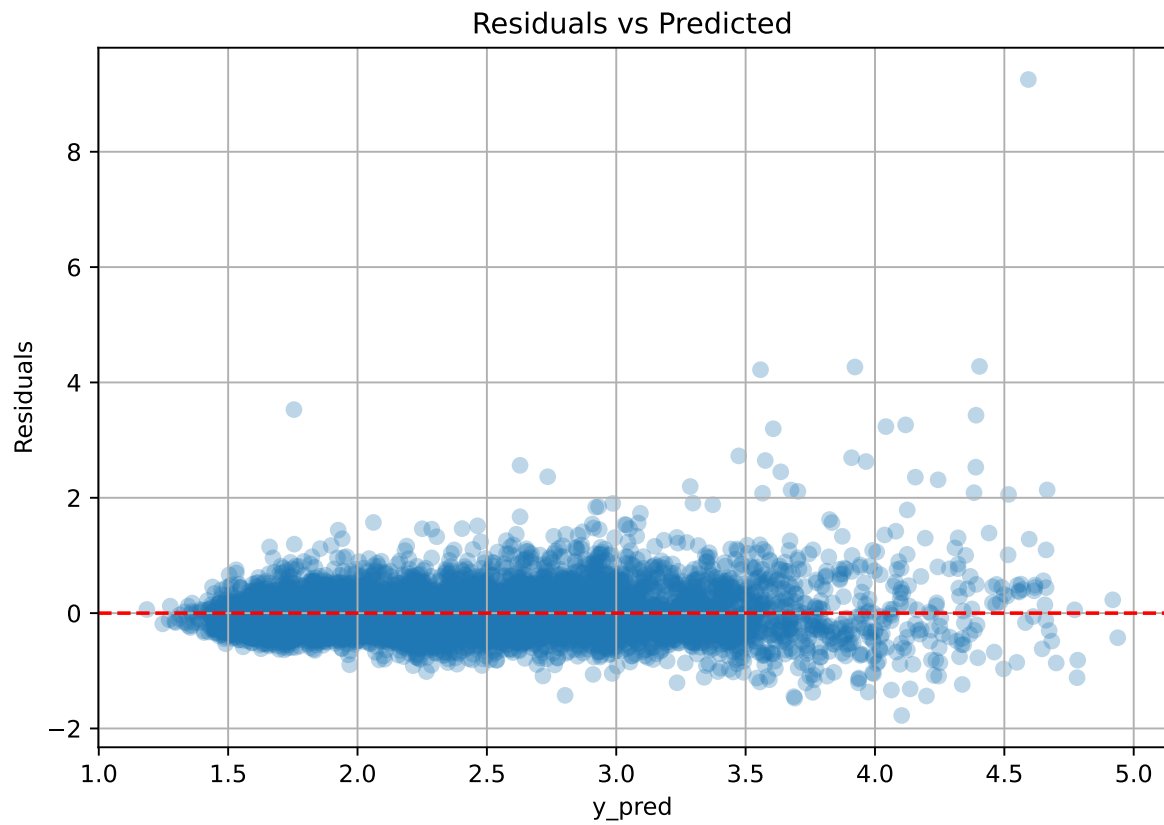
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.05394743666220058, 'reg__num_leaves': 40, 'reg__subsample': 0.7266879297446024, 'reg__colsample_bytree': 0.5414078718642084, 'reg__min_child_samples': 39, 'reg__reg_alpha': 2.538359711540569, 'reg__reg_lambda': 10.951273174809073, 'reg__min_split_gain': 0.975863803110214, 'reg__path_smooth': 3.7194800995580146, 'reg__min_child_weight': 0.053567044593560424, 'reg__feature_fraction': 0.5294864423865422, 'reg__bagging_fraction': 0.7786725035156087, 'reg__bagging_freq': 5, 'reg__max_bin': 70, 'reg__min_data_per_group': 127}

Rank 10: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.054409024982417384, 'reg__num_leaves': 56, 'reg__subsample': 0.7256369084937574, 'reg__colsample_bytree': 0.5283717901405196, 'reg__min_child_samples': 40, 'reg__reg_alpha': 3.3446933602707634, 'reg__reg_lambda': 0.9342362665479236, 'reg__min_split_gain': 0.9641831794433904, 'reg__path_smooth': 3.7664945006548853, 'reg__min_child_weight': 0.1072949977474404, 'reg__feature_fraction': 0.5277945985623973, 'reg__bagging_fraction': 0.841014956750624, 'reg__bagging_freq': 5, 'reg__max_bin': 72, 'reg__min_data_per_group': 126}

3.5.2 Scatter



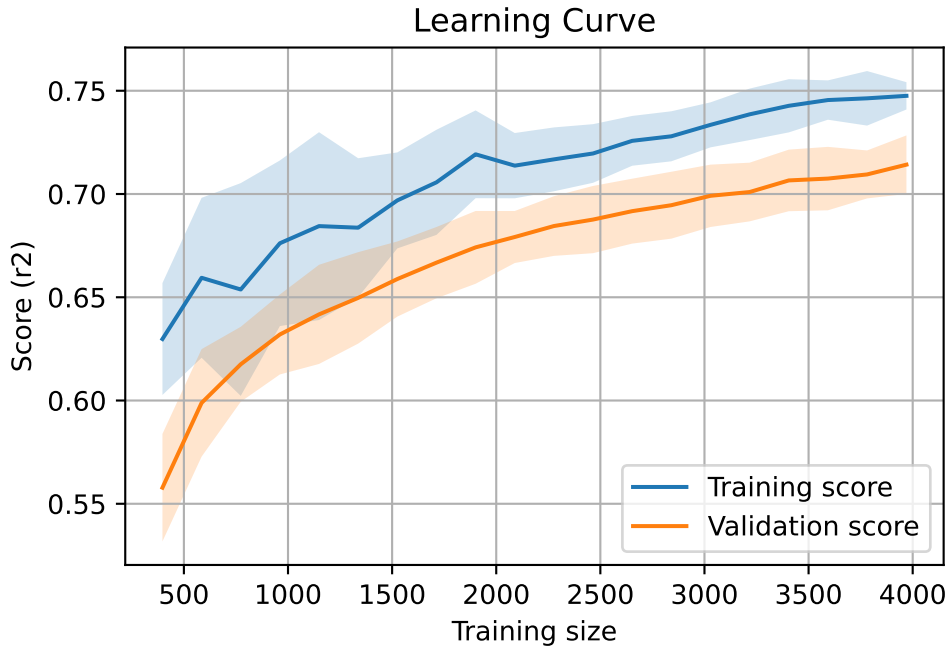
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 8104, **Len(training)** : 5673,

Len(training_real) : 3972, **Len(test_of_training)** : 1701, **Len(test)** : 2431



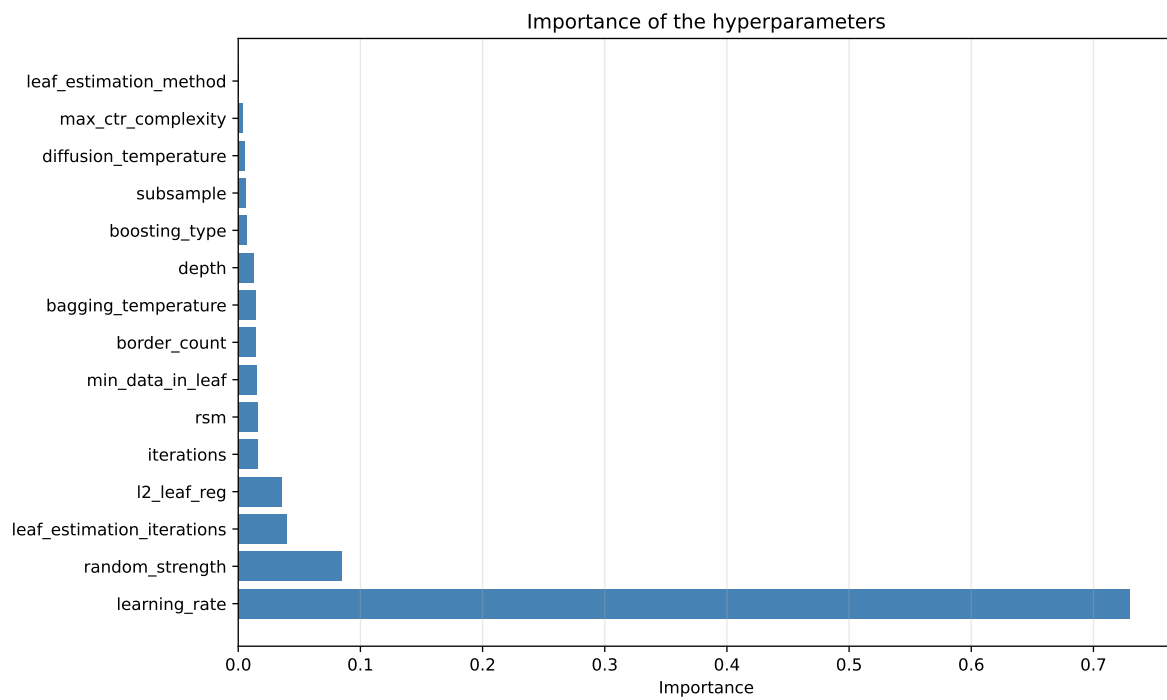
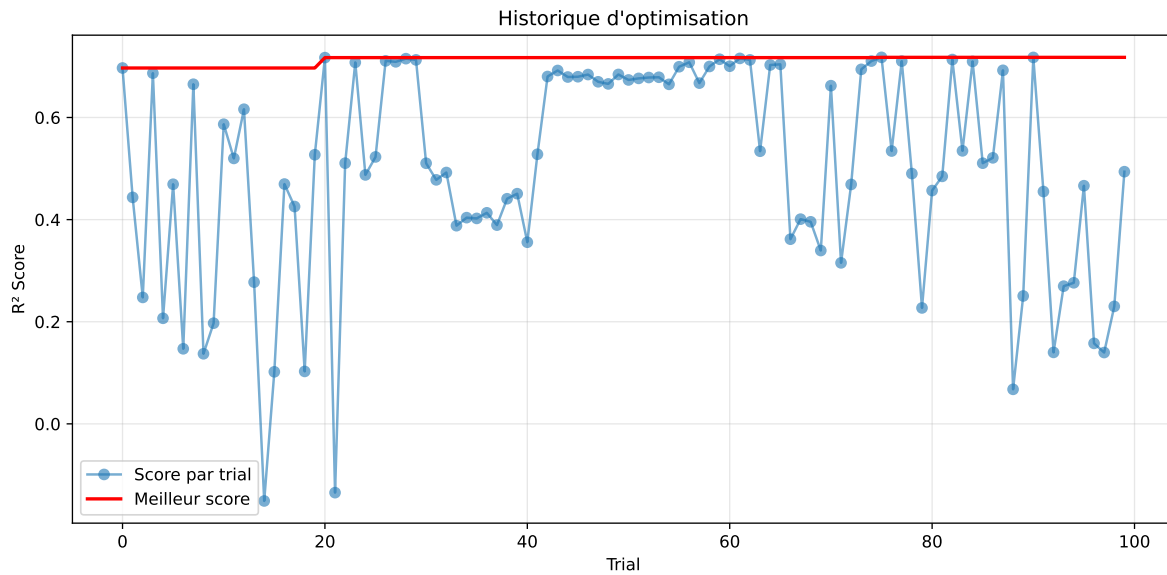
3.6 Model : CatBoost

3.6.1 Gridsearch

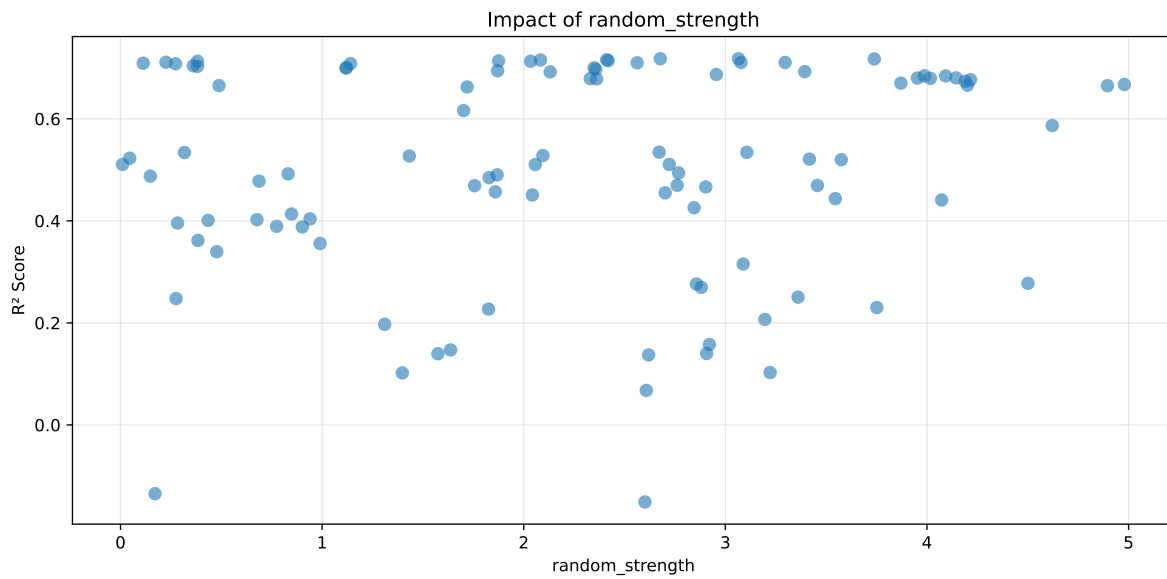
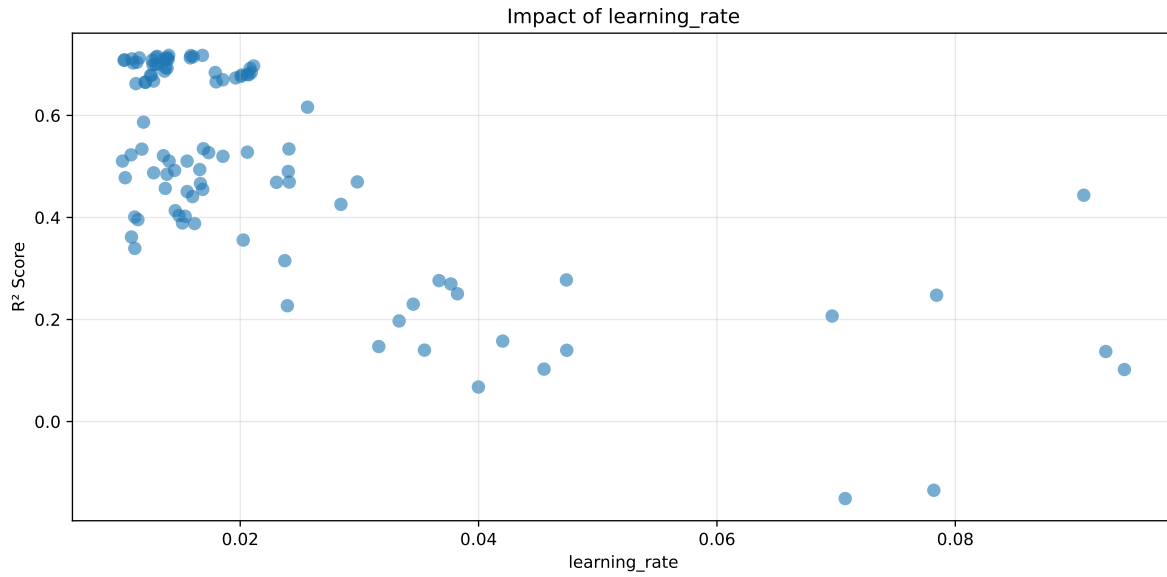
Distribution y_train vs y_test: y_train: mean=2.344, std=0.794, min=0.973, max=13.845
y_test: mean=2.367, std=0.800, min=0.959, max=8.682

===== TOP Importance FEATURES =====

feature	importance
r_umidity	20.396456
r_temperature_st_2h_std	14.521417
r_CO2	12.482644
r_temperature_st_2h_diff_last	8.320776
r_umidity_st_2h_slope	6.797642
r_NH3	6.356647
r_temperature	6.166527
r_NH3_st_2h_min	5.604826
r_NH3_st_2h_std	5.151543
r_temperature_st_2h_mean	3.127598
r_THI	3.051771
r_NH3_st_2h_slope	2.964582
r_umidity_st_2h_diff_last	1.864763
r_umidity_st_2h_std	1.610986
r_NH3_st_2h_diff_last	1.581823

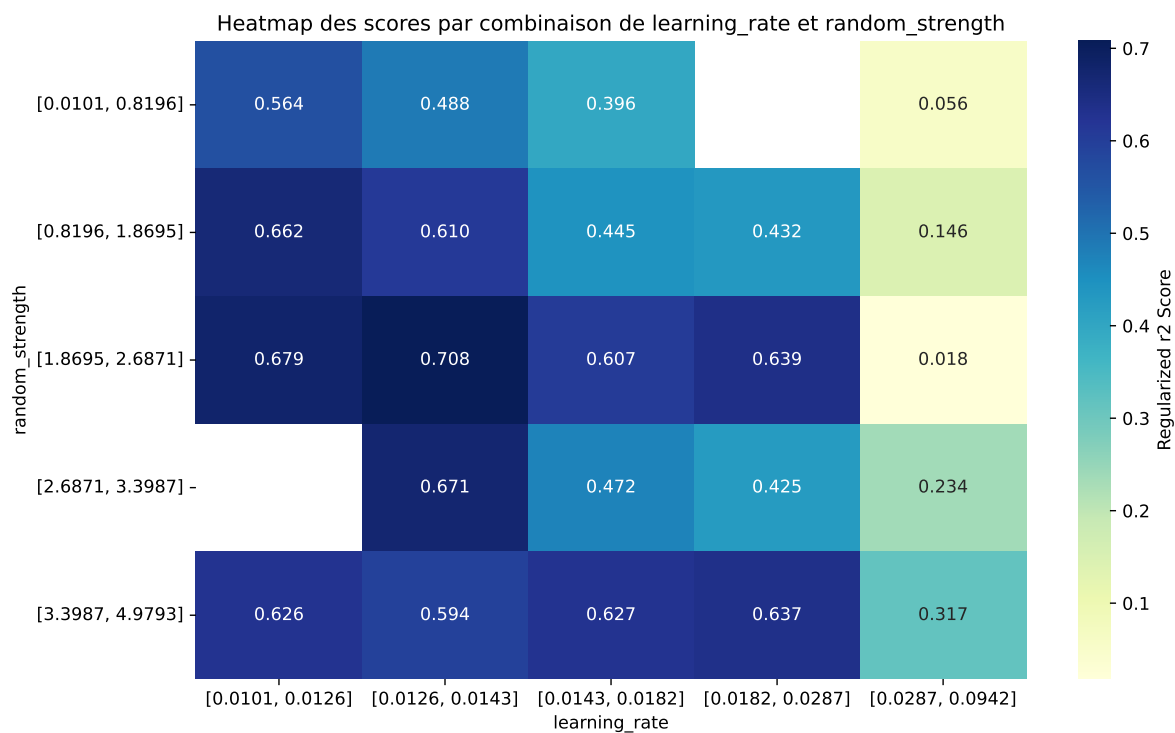


Paramètres avec >5% d'importance : ['learning_rate', 'random_strength']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7521	0.7178	0.6914	0.017	0.7178	0.6964	-0.0214	0.9703	1.0478
2	0.7511	0.7177	0.69	0.0164	0.7177	0.7004	-0.0172	0.976	1.0466
3	0.7467	0.7173	0.6883	0.0156	0.7173	0.6961	-0.0212	0.9705	1.041
4	0.7501	0.7157	0.6886	0.0142	0.7157	0.6968	-0.0189	0.9736	1.048
5	0.7454	0.7152	0.6869	0.015	0.7152	0.6908	-0.0244	0.9659	1.0422
6	0.7455	0.7141	0.6873	0.0163	0.7141	0.6919	-0.0221	0.969	1.0441
7	0.7445	0.7135	0.6863	0.0165	0.7135	0.6925	-0.0209	0.9707	1.0436
8	0.7431	0.7128	0.6849	0.0157	0.7128	0.6903	-0.0225	0.9684	1.0426
9	0.7472	0.7128	0.6828	0.0128	0.7128	0.6895	-0.0233	0.9673	1.0483
10	0.7453	0.7109	0.6812	0.013	0.7109	0.6861	-0.0248	0.9651	1.0485
---	---	---	---	---	---	---	---	---	---
100	0.9716	0.7845	0.7575	0.0144	-0.151	0.7768	-0.0077	0.9902	1.2385

Hyperparameters:

Rank 1: {'reg__iterations': 900, 'reg__depth': 5, 'reg__learning_rate': 0.01400171537173301, 'reg__l2_leaf_reg': 4.1934864873413105, 'reg__bagging_temperature': 0.2406571279078027, 'reg__random_strength': 3.065220827604029, 'reg__subsample': 0.6652579360728419, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8730029957886317, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2641.270135837555}

Rank 2: {'reg__iterations': 800, 'reg__depth': 6, 'reg__learning_rate': 0.016829805682413677, 'reg__l2_leaf_reg': 49.92567572222231, 'reg__bagging_temperature': 0.13455834751320372, 'reg__random_strength': 2.6774622500255085, 'reg__subsample': 0.6873408508813378, 'reg__min_data_in_leaf': 48, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6577451728767197, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3203.7005507346785}

Rank 3: {'reg__iterations': 1000, 'reg__depth': 5, 'reg__learning_rate': 0.0158564670842744, 'reg__l2_leaf_reg': 44.65240153533869, 'reg__bagging_temperature': 1.9335270503370987, 'reg__random_strength': 3.73880205714993, 'reg__subsample': 0.8103516676237477, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7258396545707487, 'reg__border_count': 61, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1567.3901805772762}

Rank 4: {'reg__iterations': 900, 'reg__depth': 5, 'reg__learning_rate': 0.013052367562689338, 'reg__l2_leaf_reg': 15.03992515937583, 'reg__bagging_temperature': 3.817290220214362, 'reg__random_strength': 2.4115274928498285, 'reg__subsample': 0.8006786629903371, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8044441418436747, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4097.693120546348}

Rank 5: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.016055620742781676, 'reg__l2_leaf_reg': 20.799089540762903, 'reg__bagging_temperature': 3.2922828184981556, 'reg__random_strength': 2.0833780475513657, 'reg__subsample': 0.7332228044621624, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.7574470612243421, 'reg__border_count': 102, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7460.650836587622}

Rank 6: {'reg__iterations': 900, 'reg__depth': 5, 'reg__learning_rate': 0.012921402945825078, 'reg__l2_leaf_reg': 14.77820410659104, 'reg__bagging_temperature': 3.6614938933771817, 'reg__random_strength': 2.4191615030241946, 'reg__subsample': 0.8074628533663928, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.8055592199664936, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4247.49546780922}

Rank 7: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.013890733338423404, 'reg__l2_leaf_reg': 21.24762787910417, 'reg__bagging_temperature': 3.427748151520807, 'reg__random_strength': 1.8753457267965288, 'reg__subsample': 0.8836429612221286,

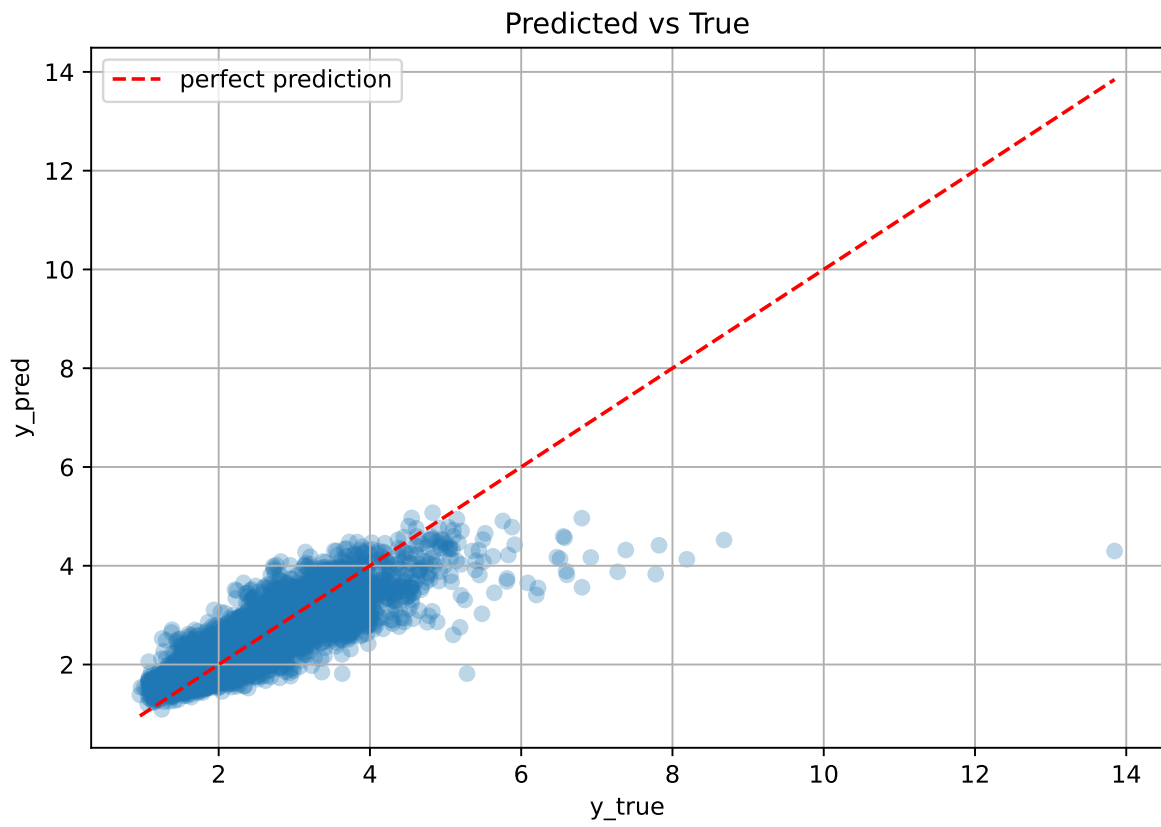
'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8613912062737313, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 9599.325252995508}

Rank 8: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.015815704337822287, 'reg__l2_leaf_reg': 22.935839679710735, 'reg__bagging_temperature': 3.410247223439139, 'reg__random_strength': 2.034188403775492, 'reg__subsample': 0.733621072649116, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.767634643379333, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6933.094475033969}

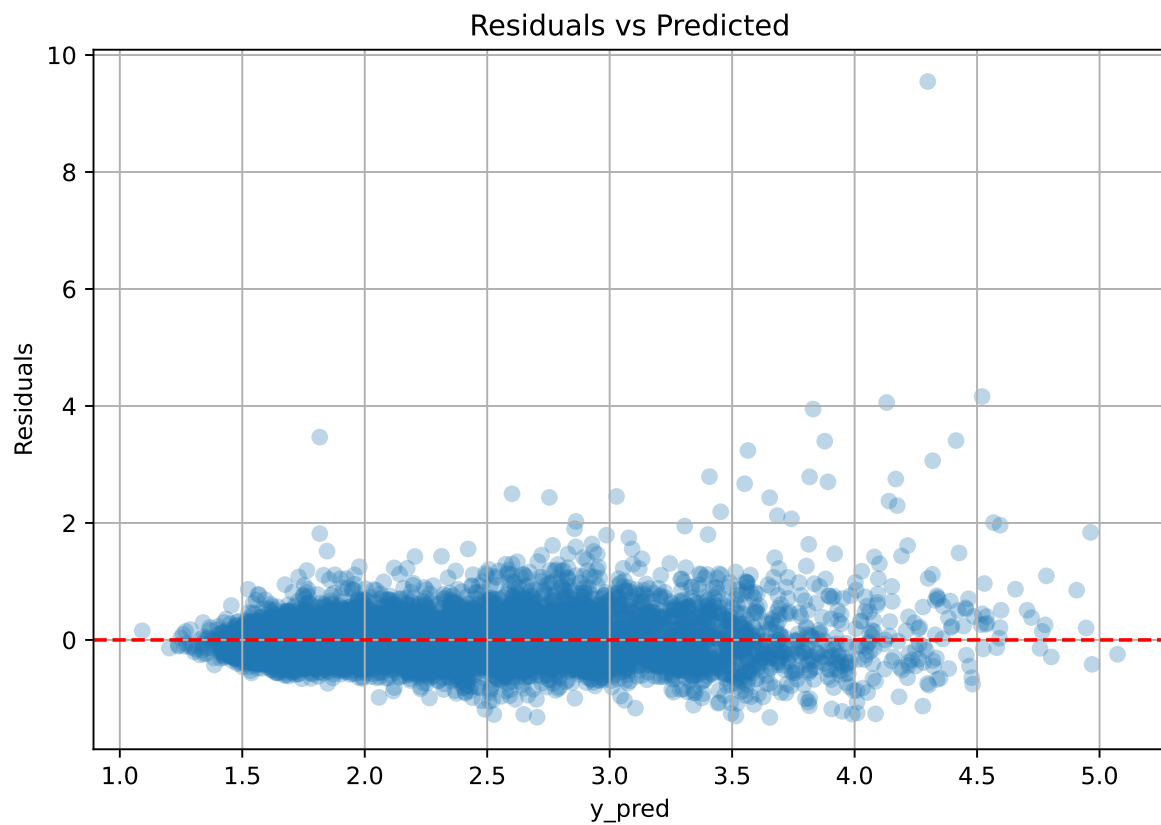
Rank 9: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.011533629763879377, 'reg__l2_leaf_reg': 15.643752302159823, 'reg__bagging_temperature': 3.646852549980908, 'reg__random_strength': 0.38224279491301405, 'reg__subsample': 0.7982907644010405, 'reg__min_data_in_leaf': 13, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.8000543122850096, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8899.534062733288}

Rank 10: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.010900936141561795, 'reg__l2_leaf_reg': 22.299121847901525, 'reg__bagging_temperature': 0.23701573931557268, 'reg__random_strength': 0.22539931154260806, 'reg__subsample': 0.7301451273667469, 'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7819856055580869, 'reg__border_count': 102, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7507.793833425595}

3.6.2 Scatter



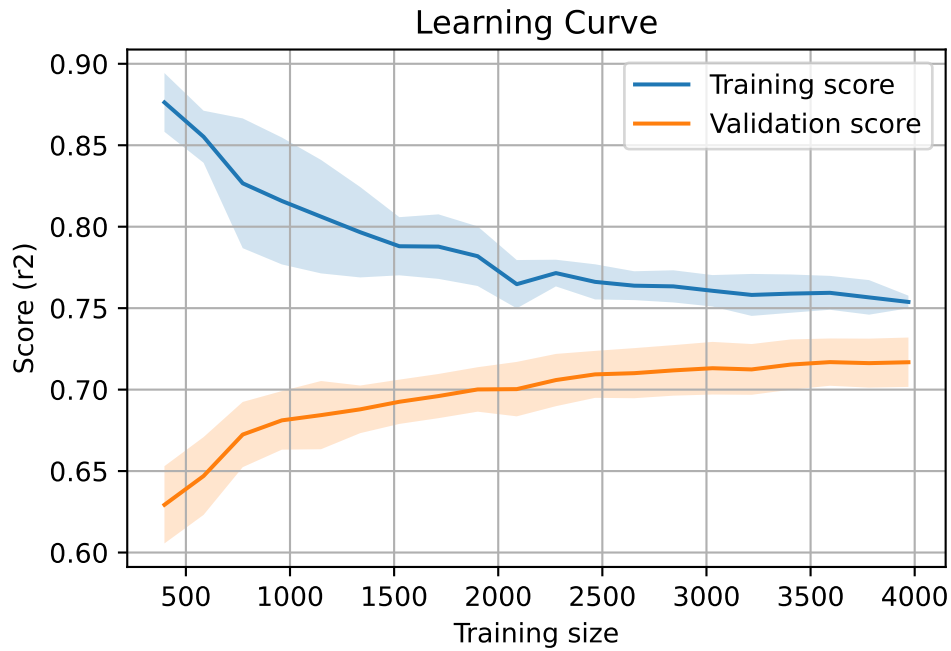
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 8104, **Len(training)** : 5673,

Len(training_real) : 3972, **Len(test_of_training)** : 1701, **Len(test)** : 2431



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.545	0.564	0.015	0.537	0.537	-0.027	0.952	0.965	nan
LGBM	0.748	0.713	0.016	0.678	0.694	-0.019	0.973	1.049	0.7128
CatBoost	0.752	0.718	0.017	0.691	0.696	-0.021	0.97	1.048	0.7178

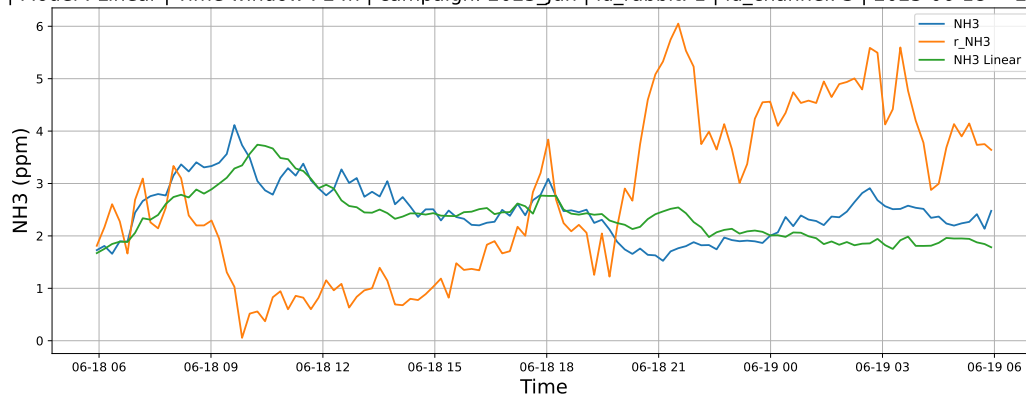
4 Plot

4.1 Standalone plots

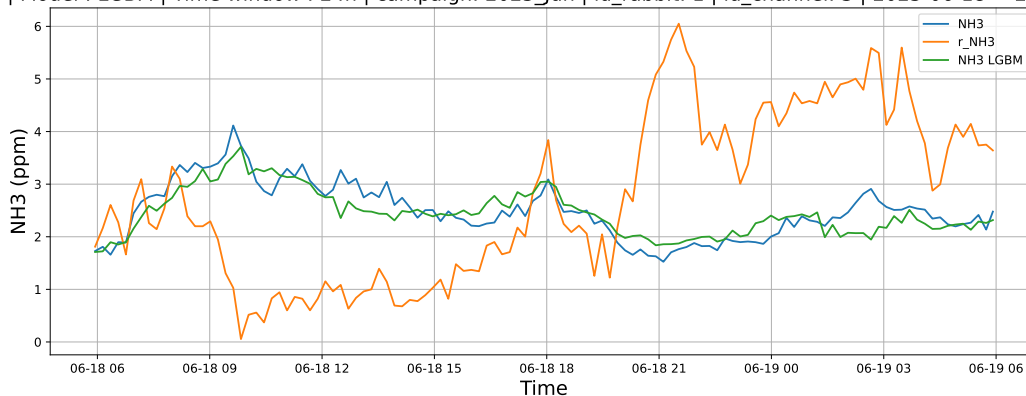
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

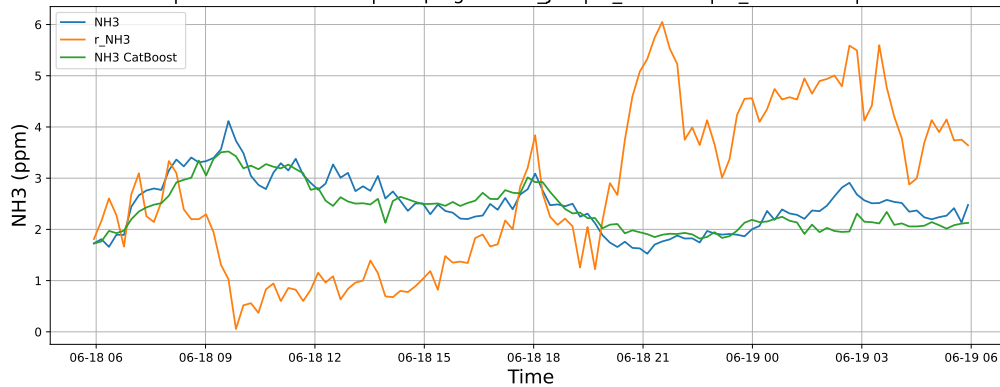
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

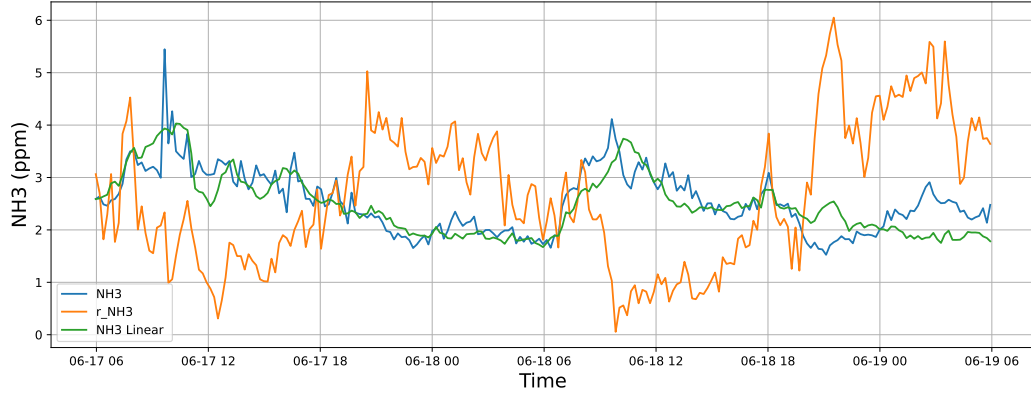


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

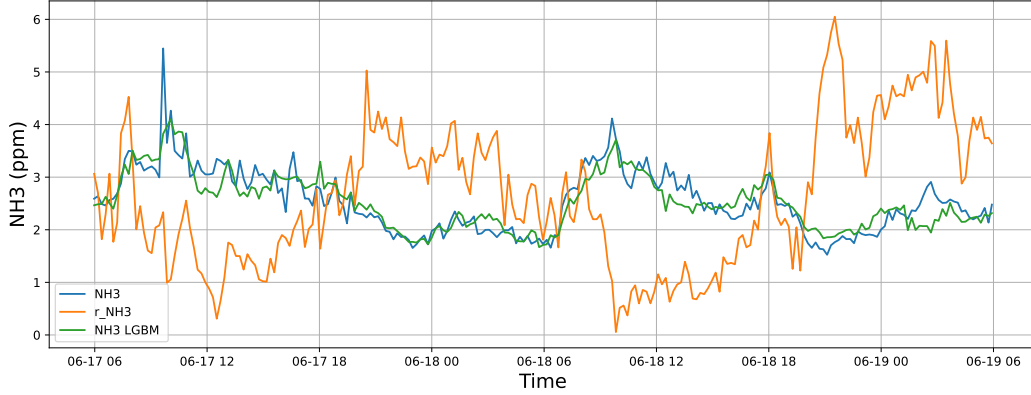


4.1.1.2 Time window : 48

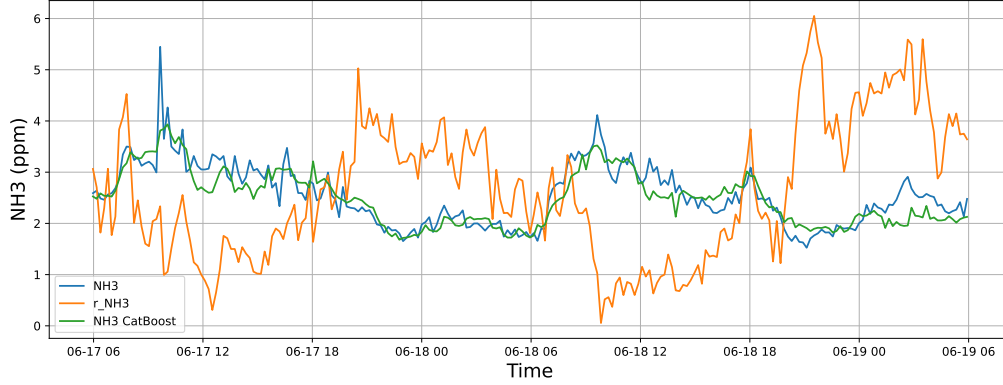
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

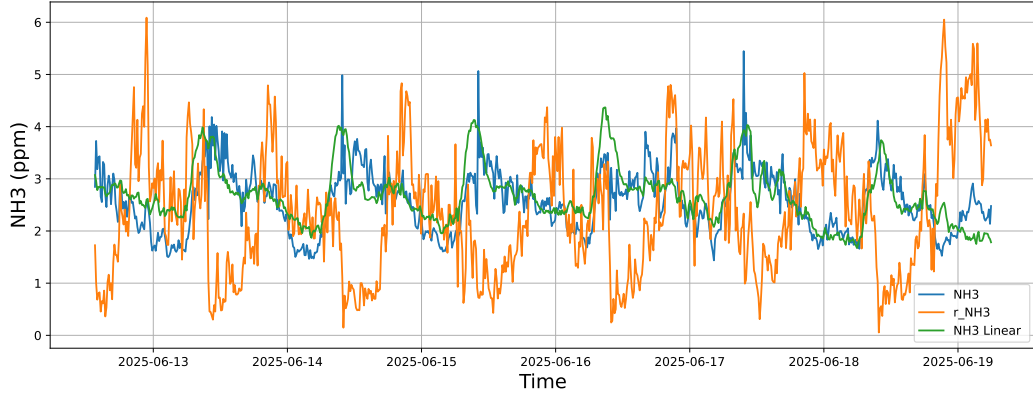


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

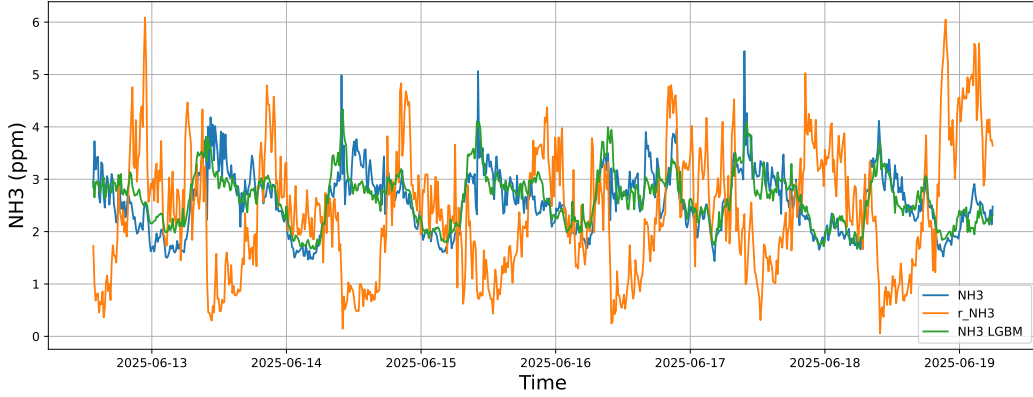


4.1.1.3 Time window : ALL

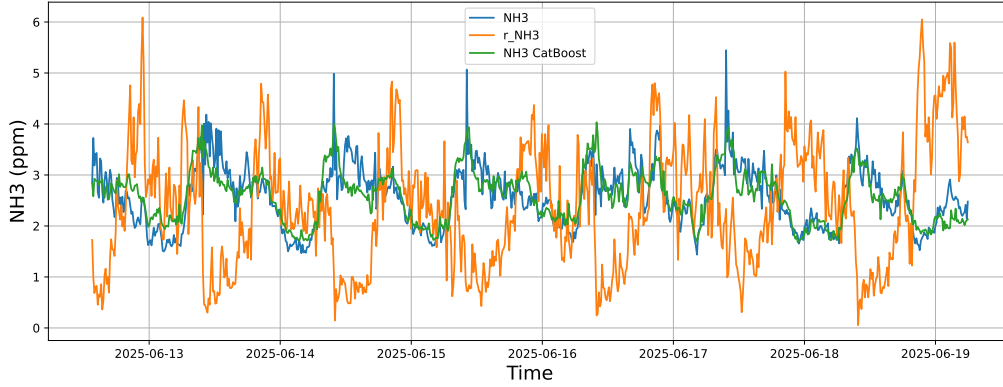
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



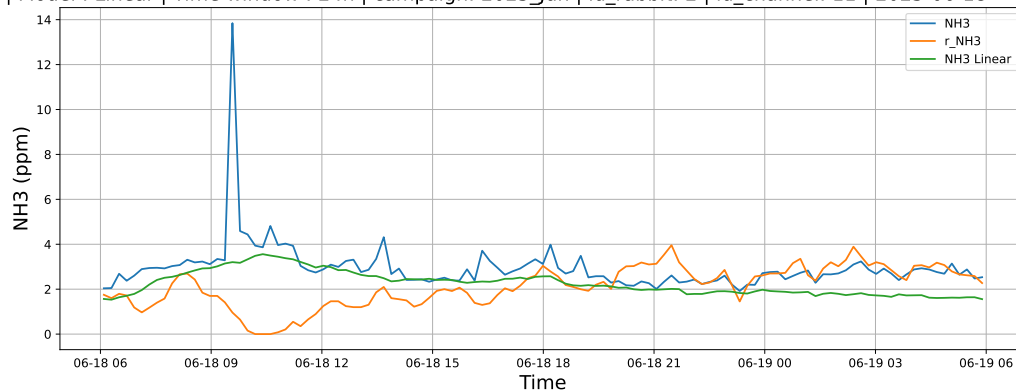
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



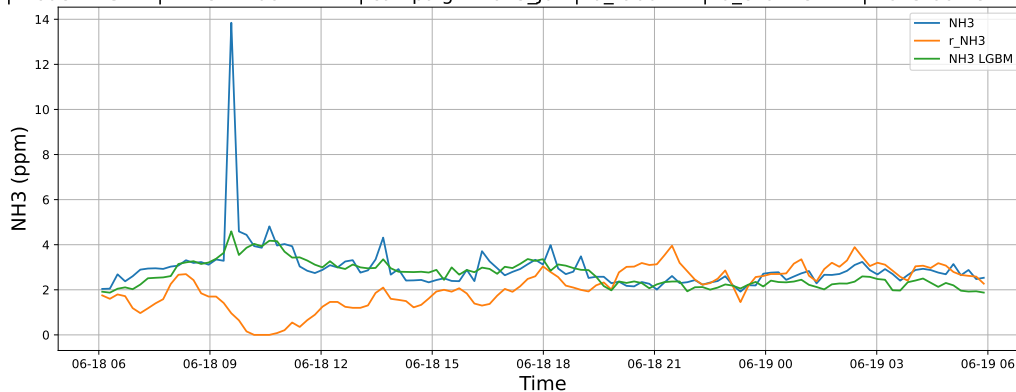
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

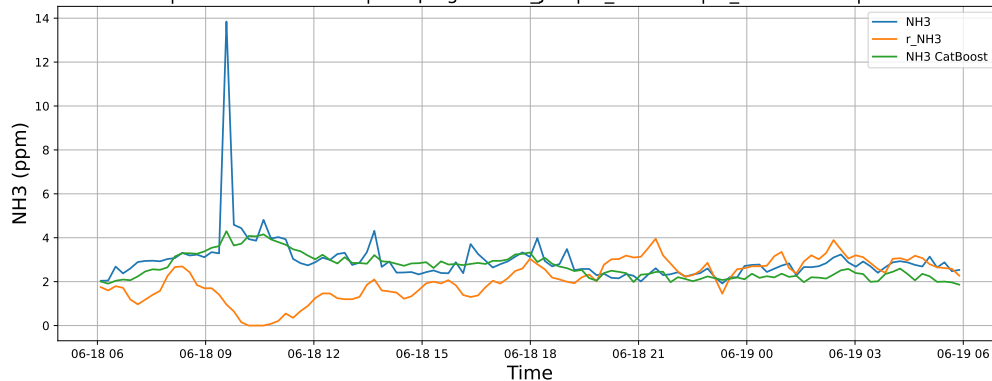
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

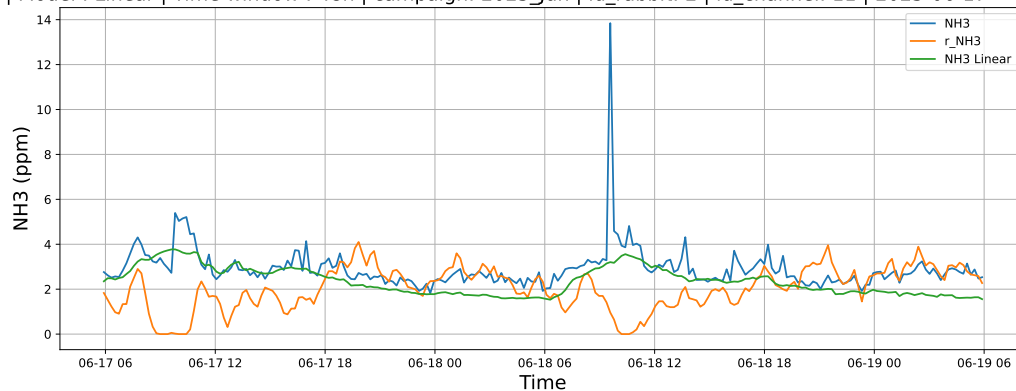


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

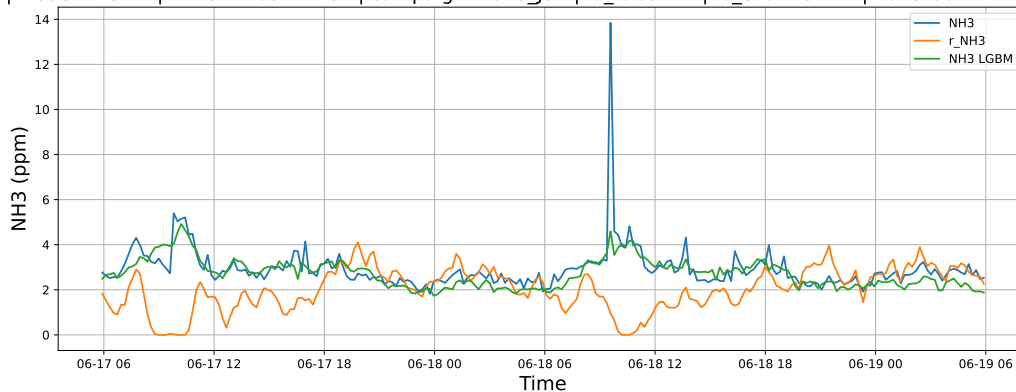


4.1.2.2 Time window : 48

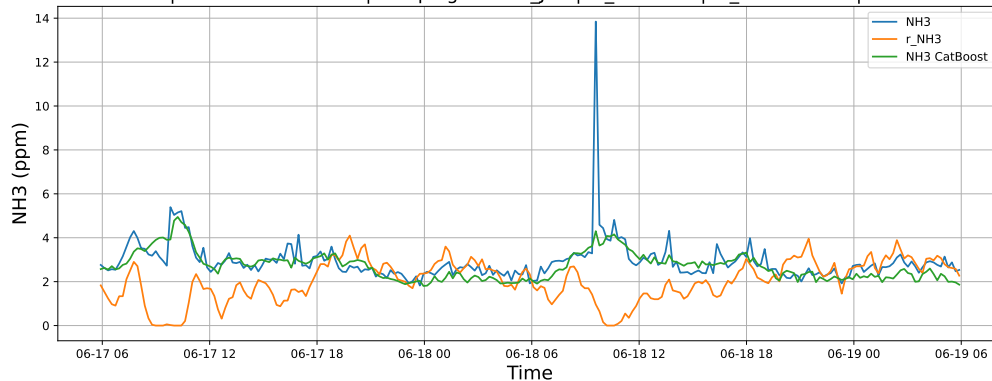
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

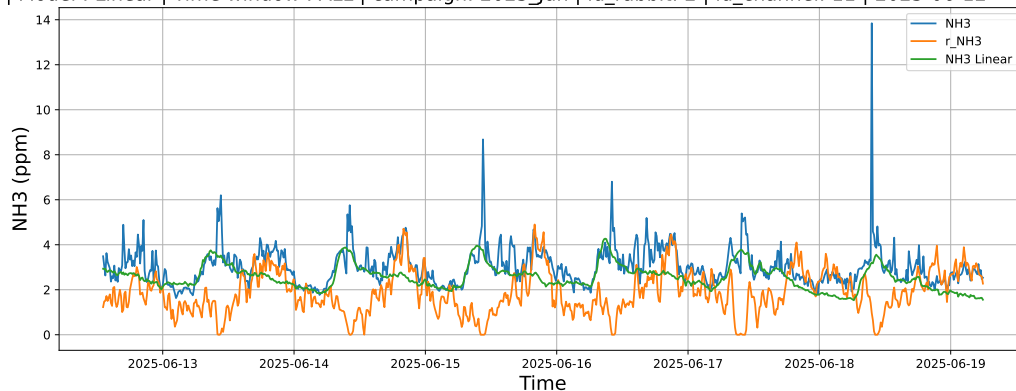


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

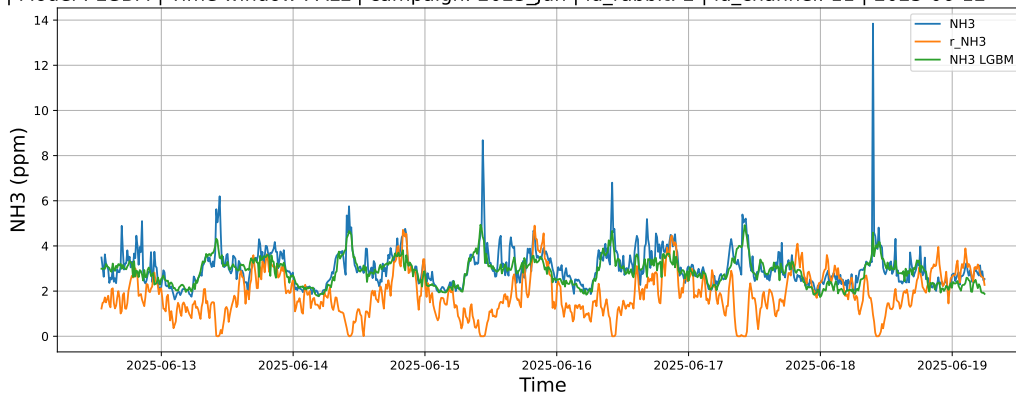


4.1.2.3 Time window : ALL

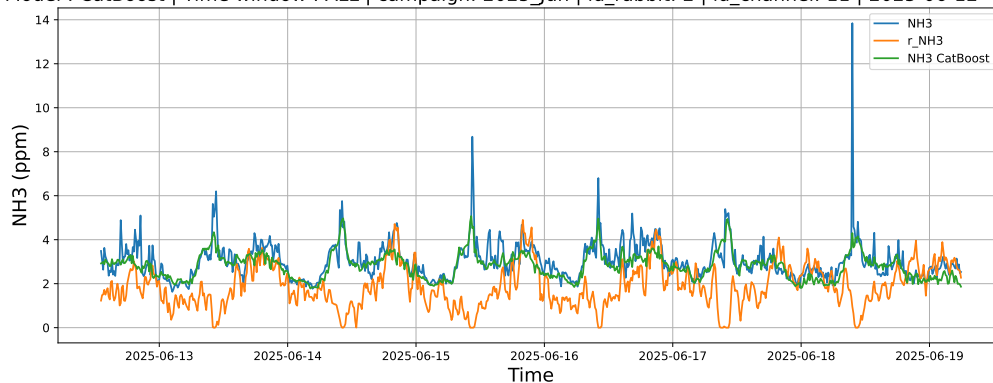
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



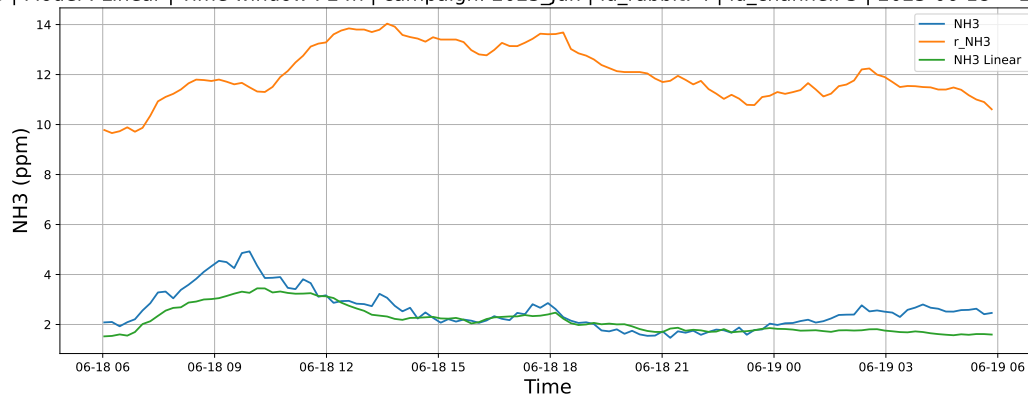
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



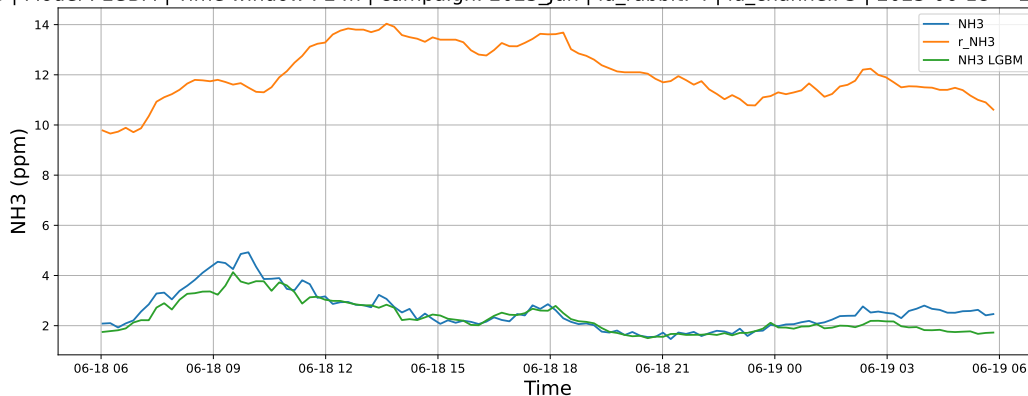
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

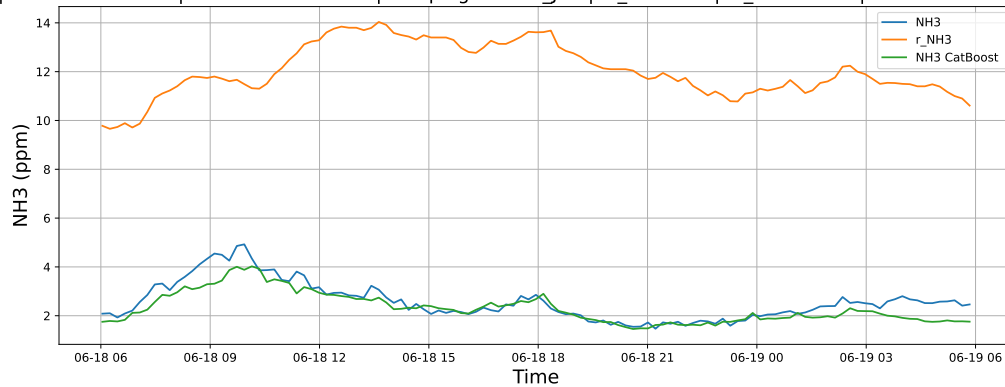
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

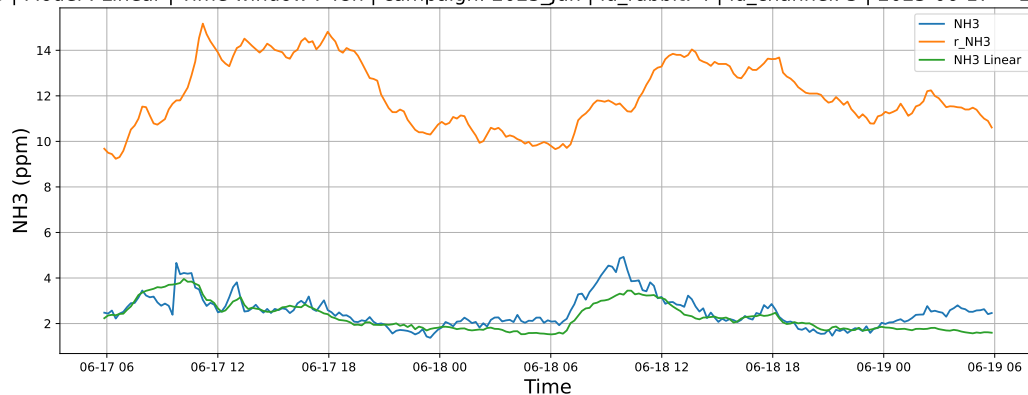


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

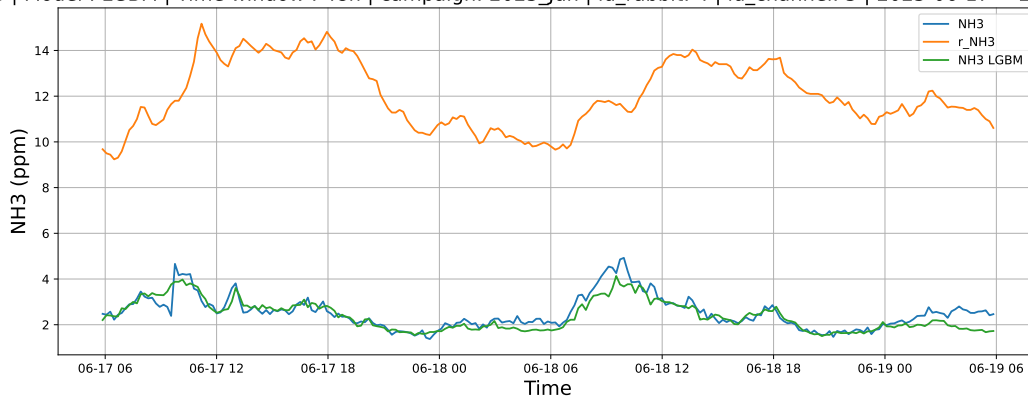


4.1.3.2 Time window : 48

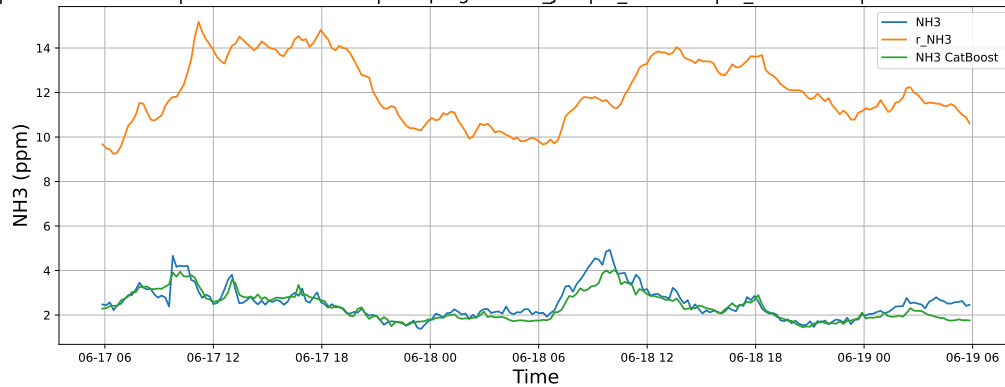
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

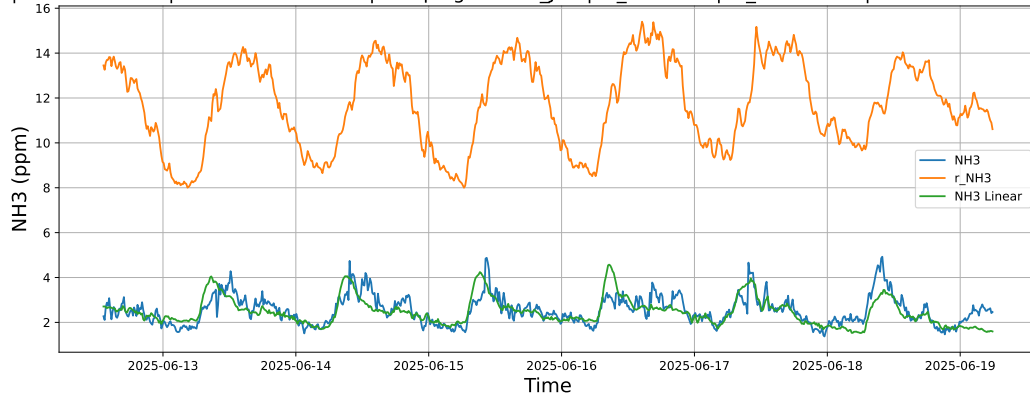


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

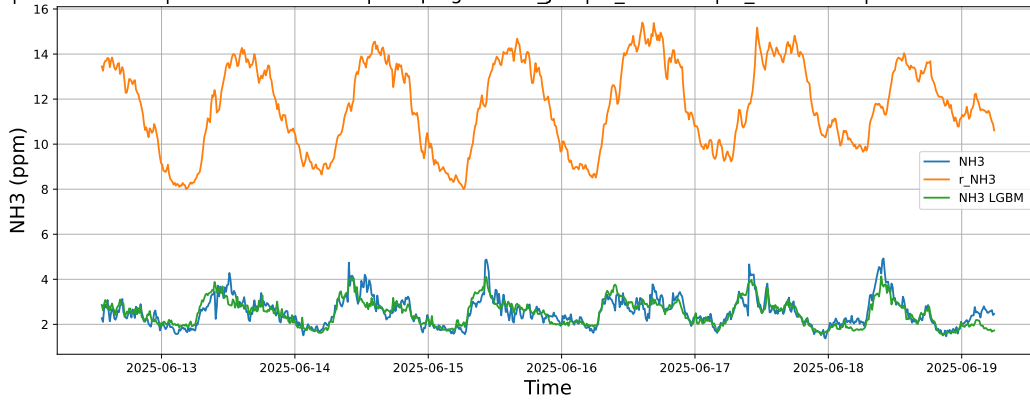


4.1.3.3 Time window : ALL

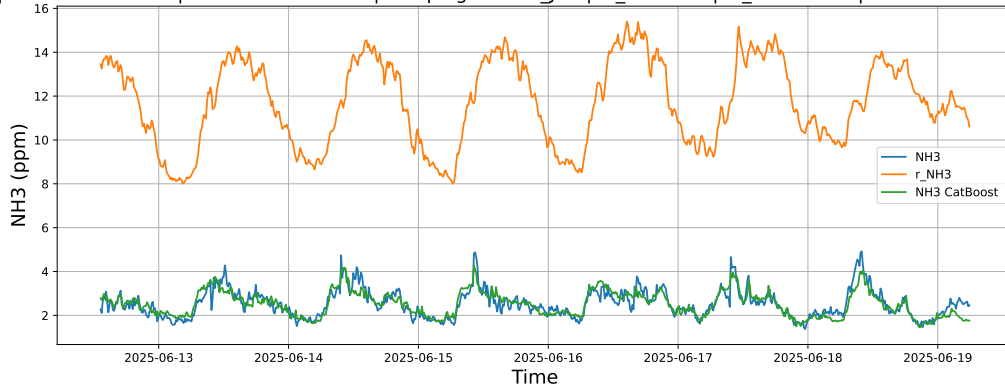
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



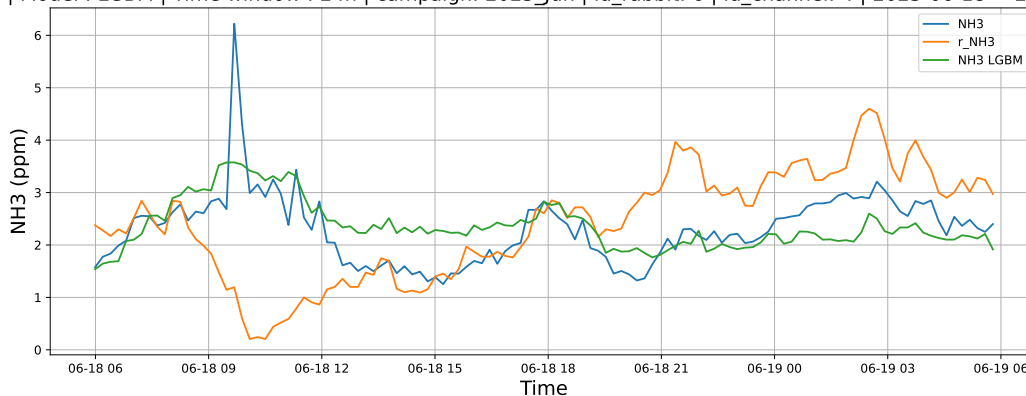
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

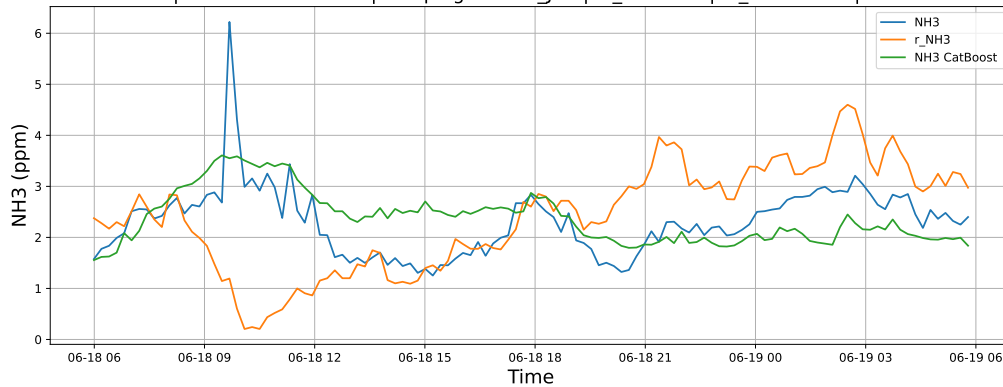
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

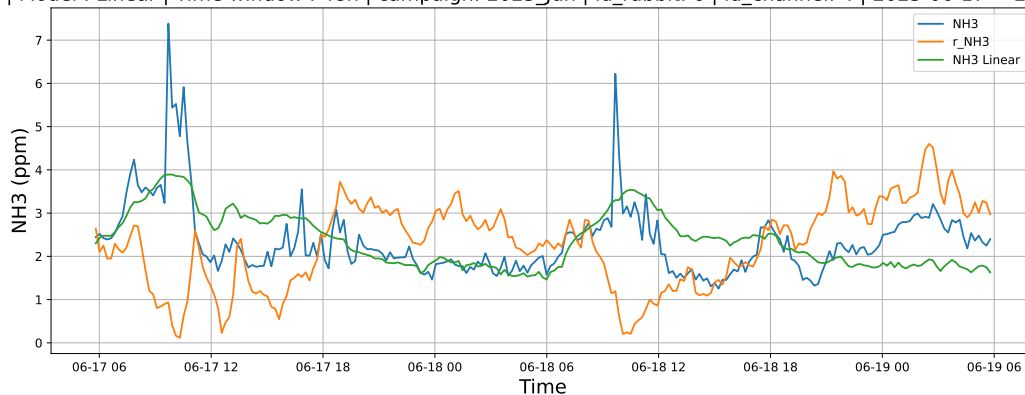


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

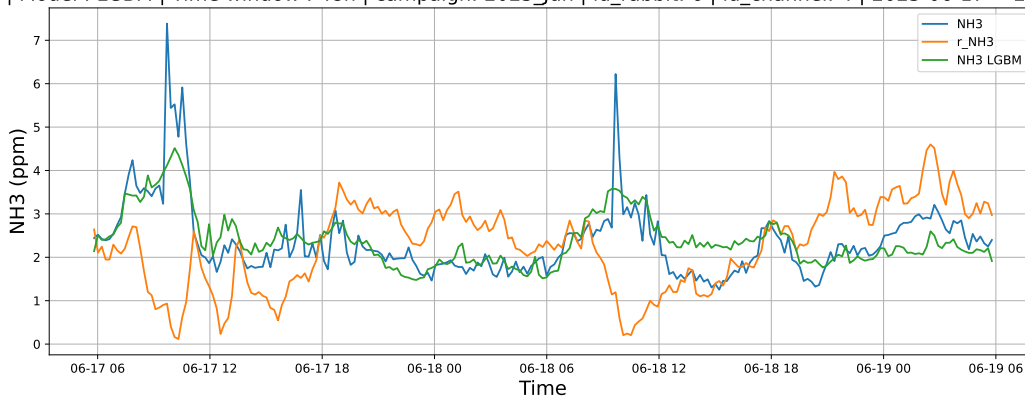


4.1.4.2 Time window : 48

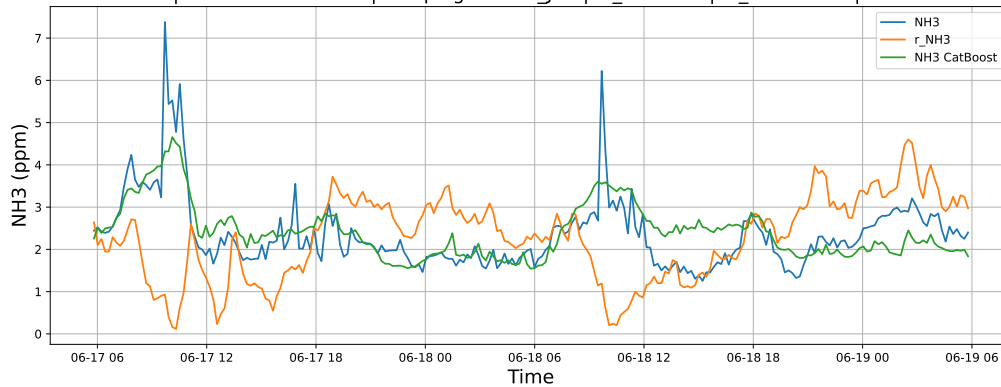
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

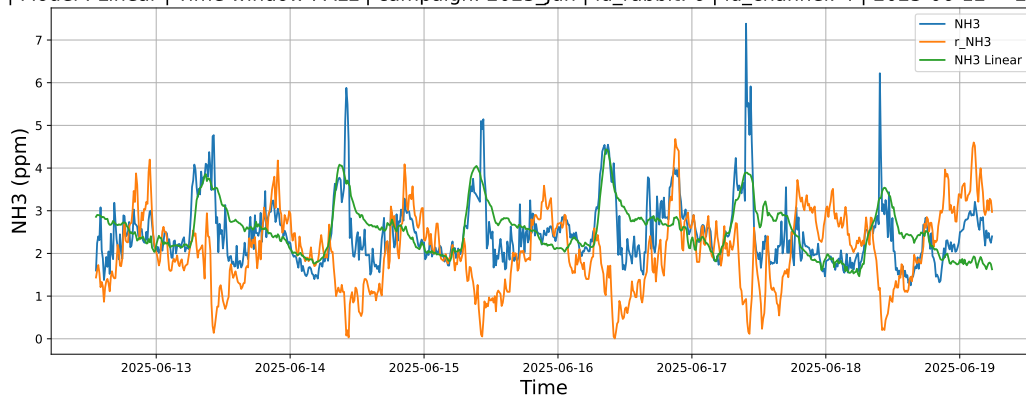


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

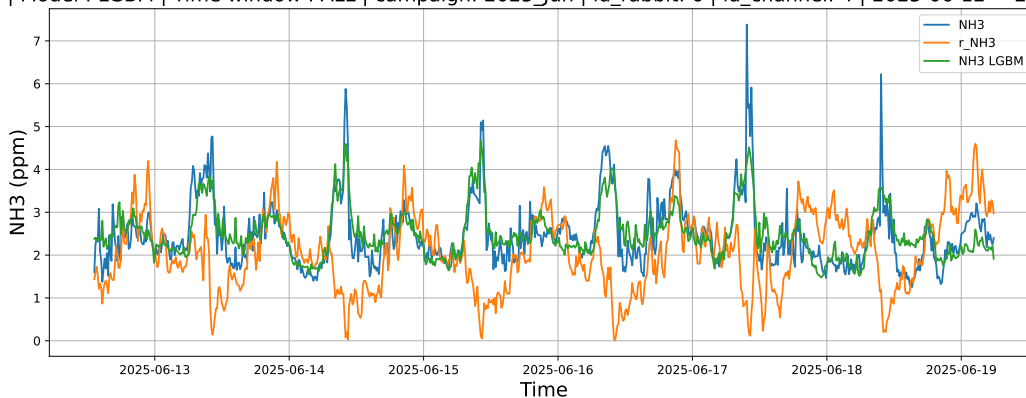


4.1.4.3 Time window : ALL

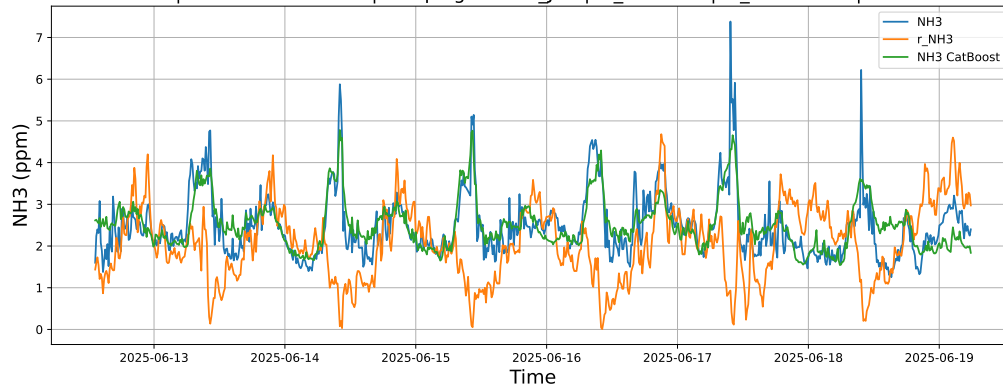
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



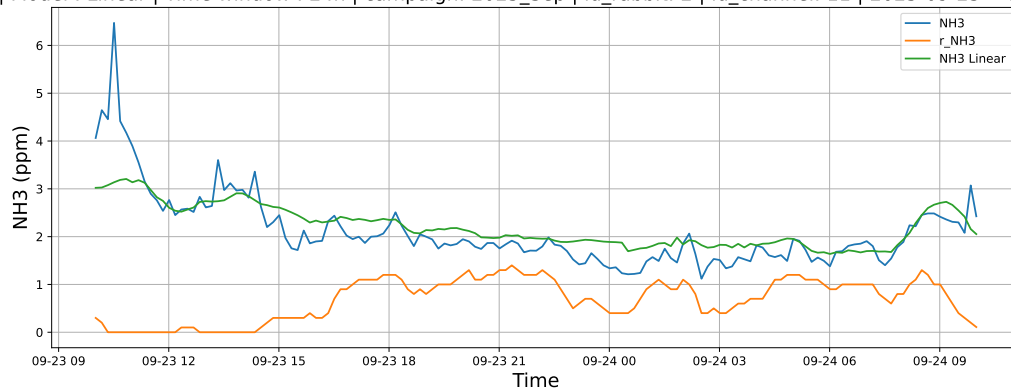
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



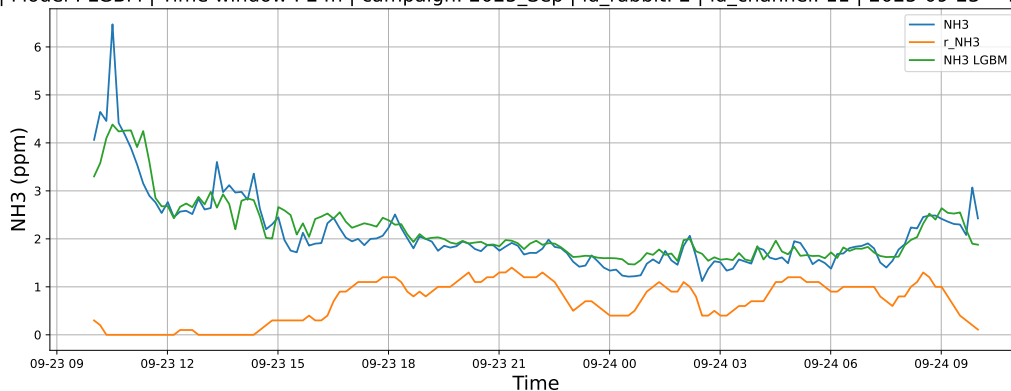
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

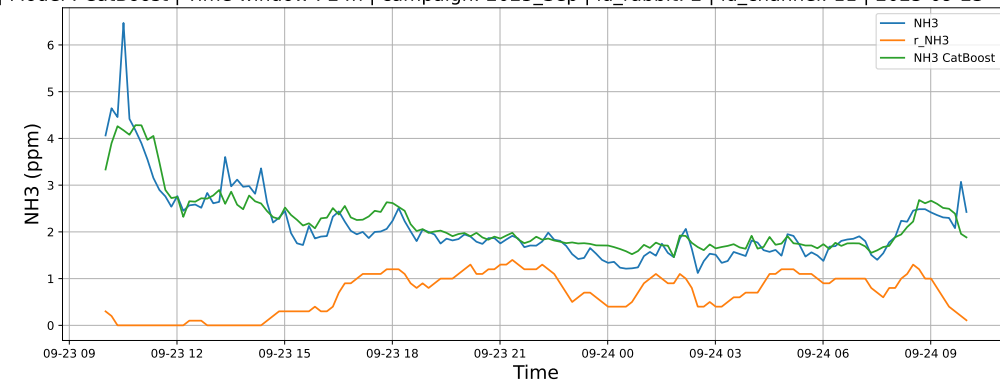
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

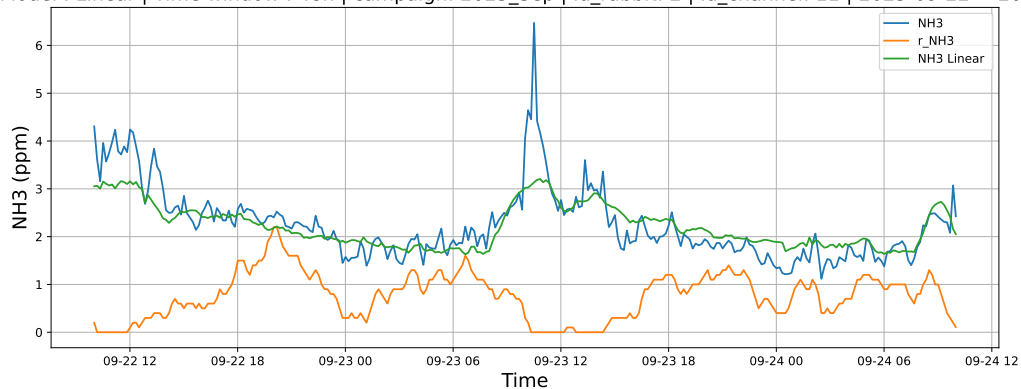


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

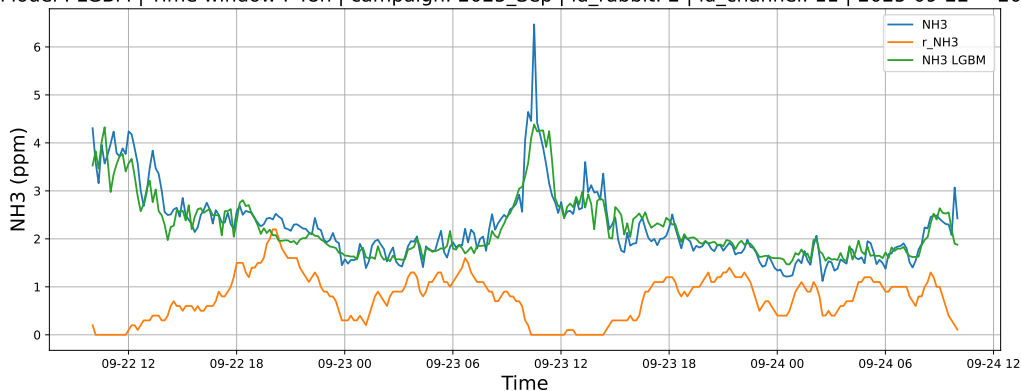


4.1.5.2 Time window : 48

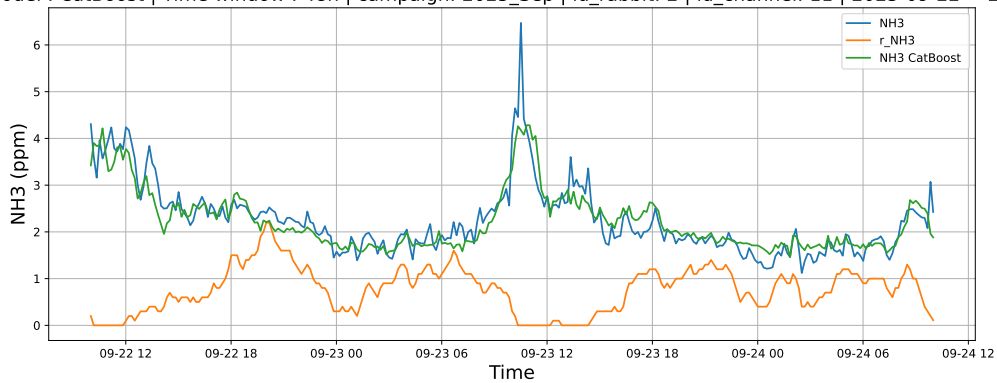
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

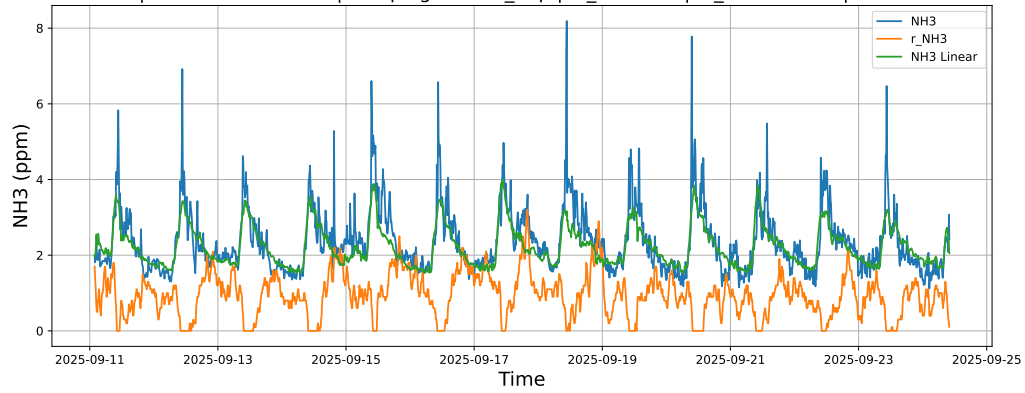


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

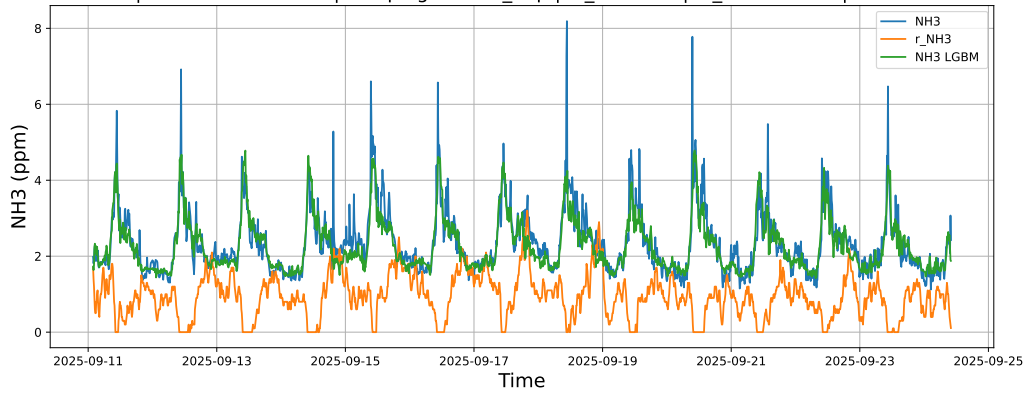


4.1.5.3 Time window : ALL

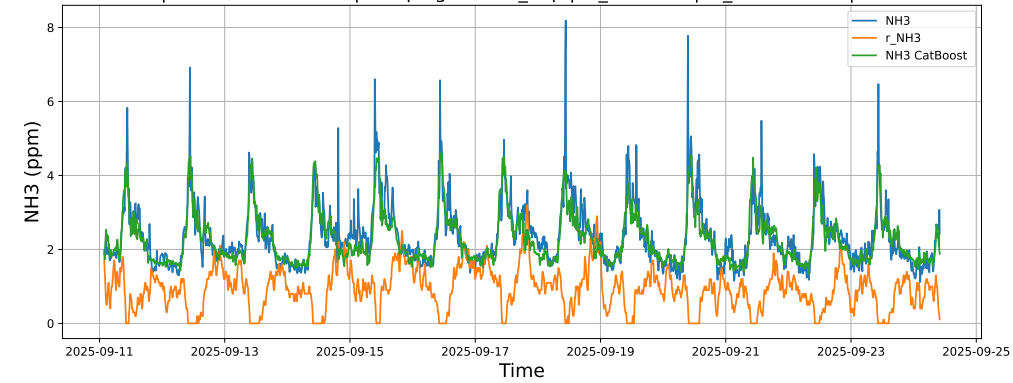
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



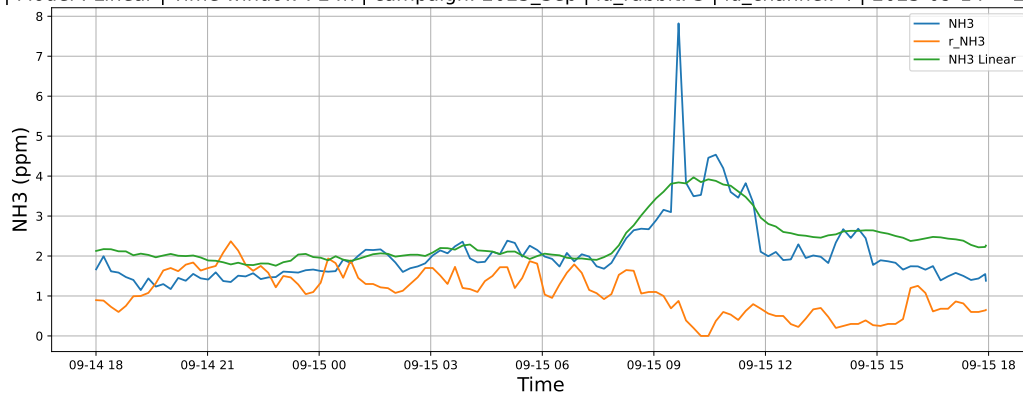
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



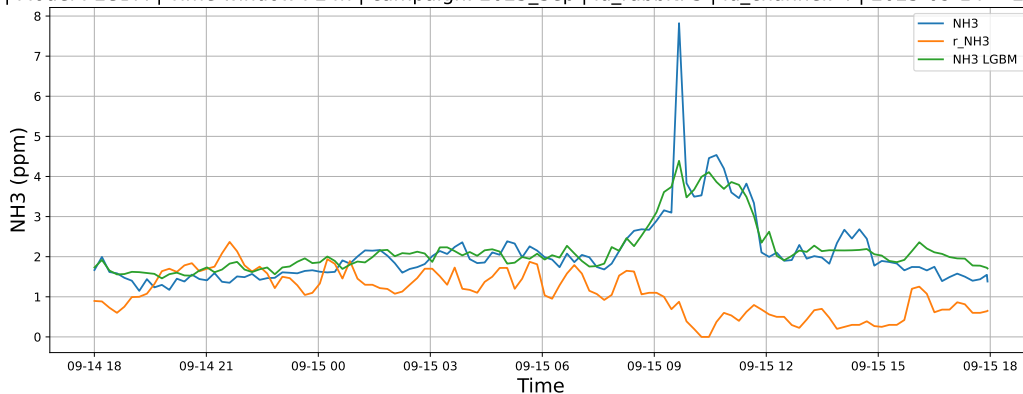
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

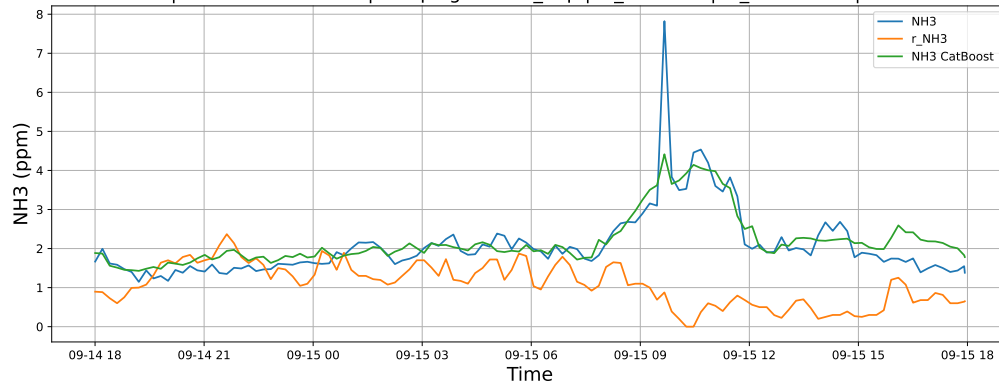
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

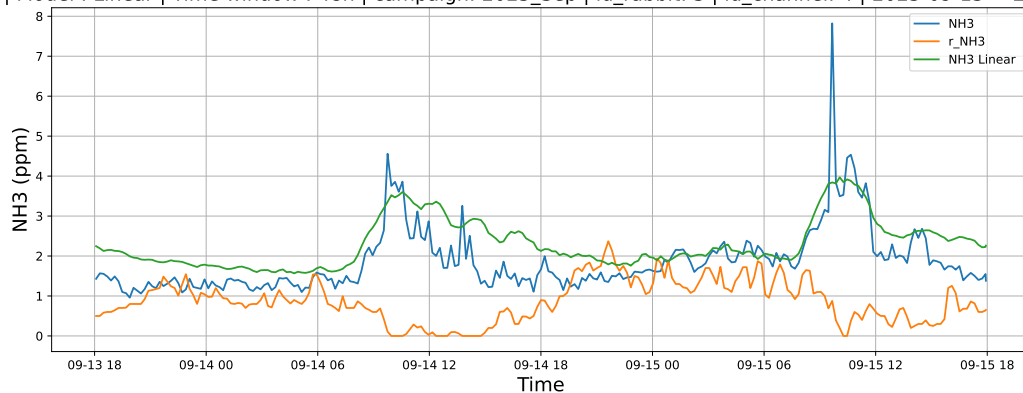


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

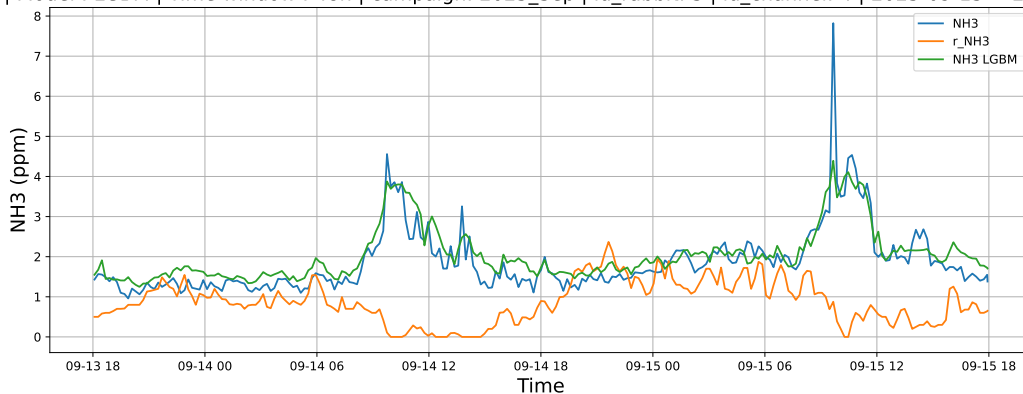


4.1.6.2 Time window : 48

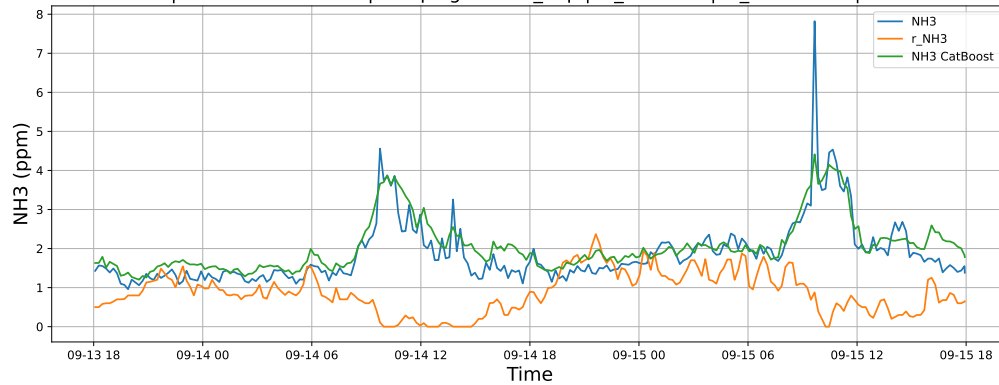
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

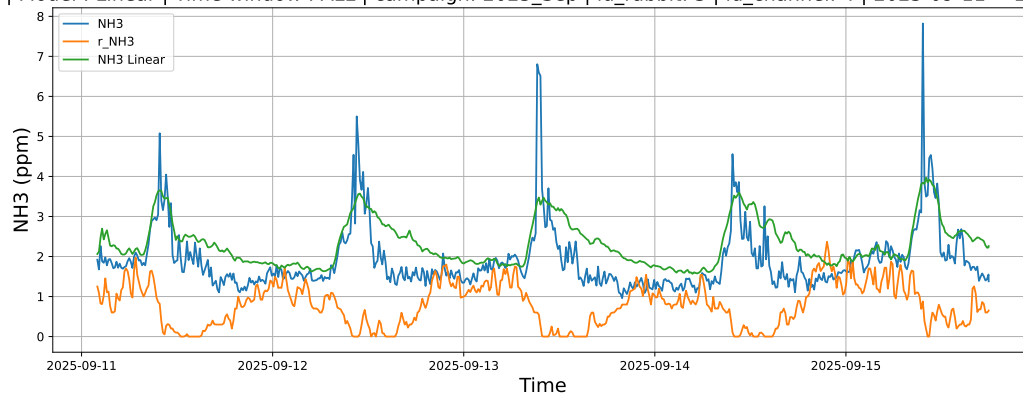


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

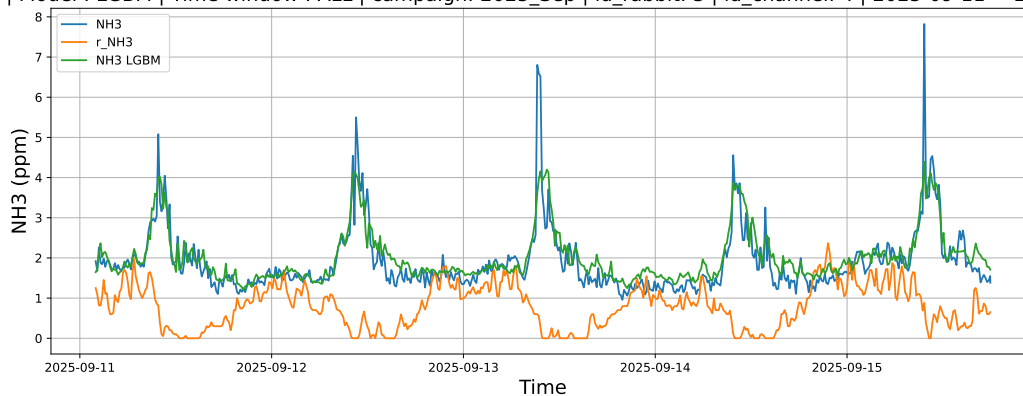


4.1.6.3 Time window : ALL

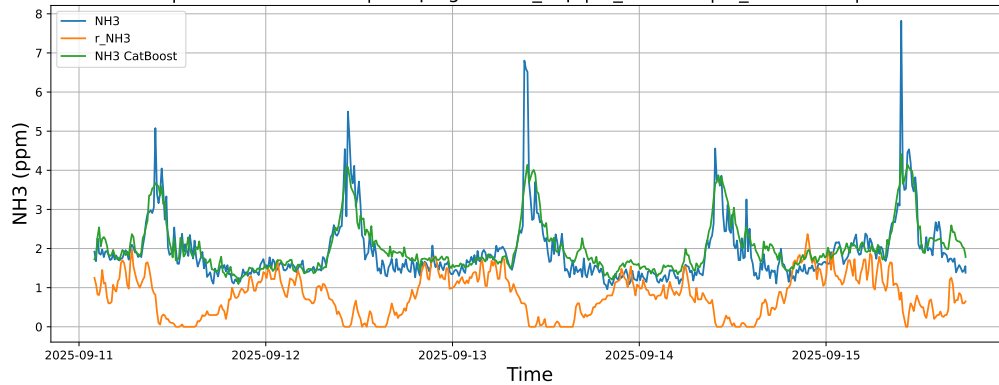
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



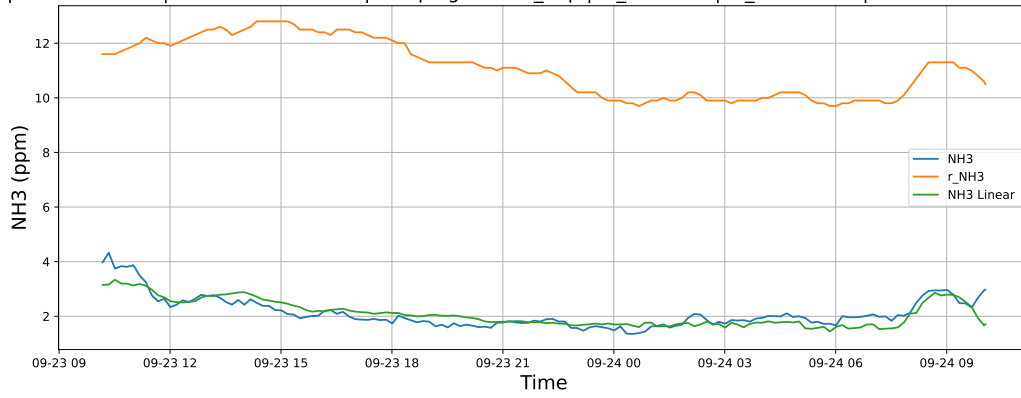
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



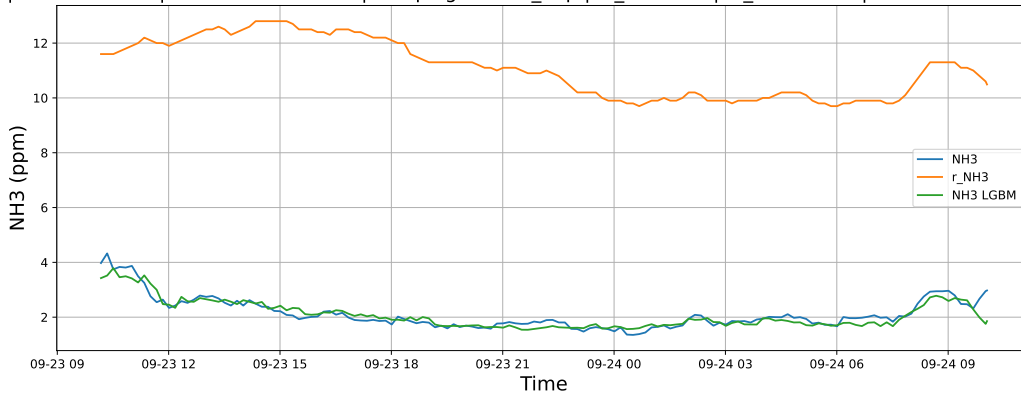
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

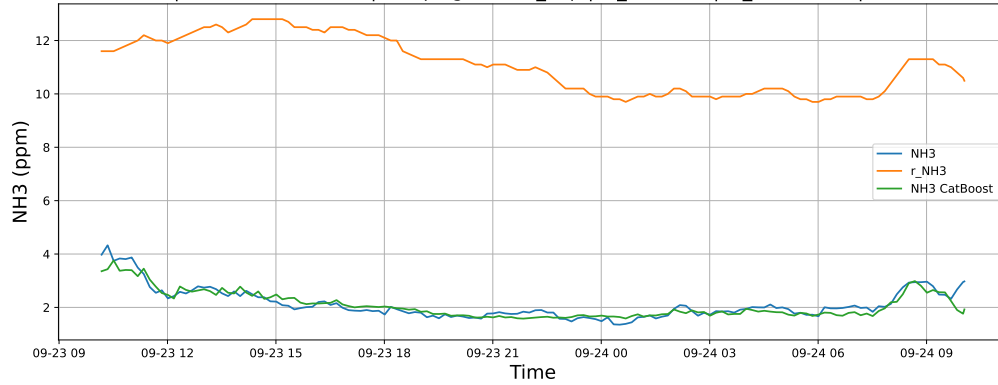
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

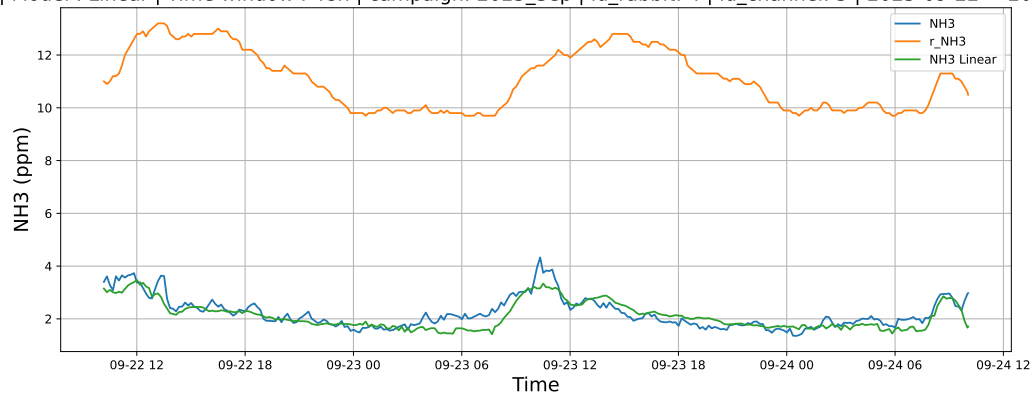


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

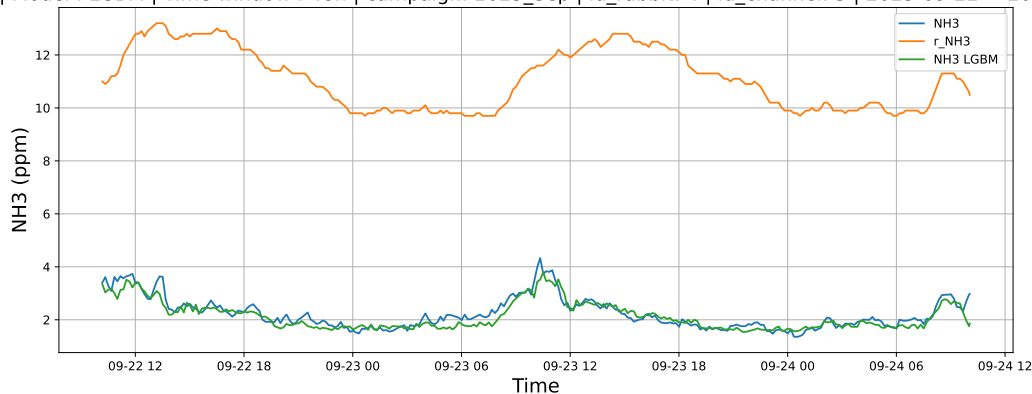


4.1.7.2 Time window : 48

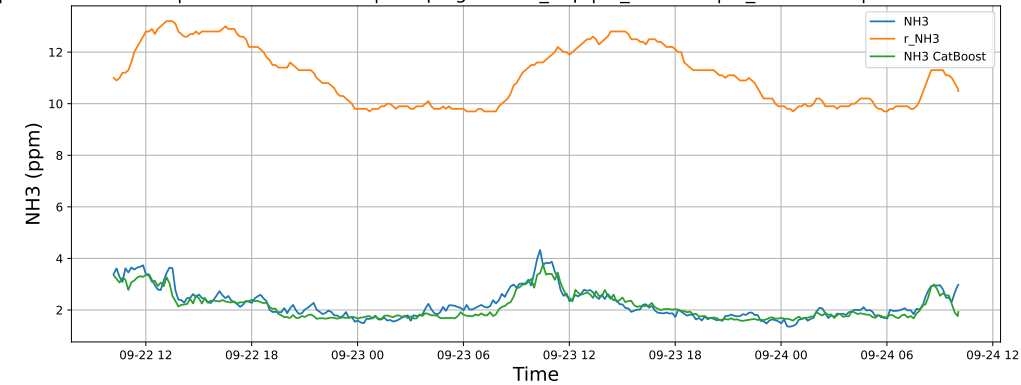
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

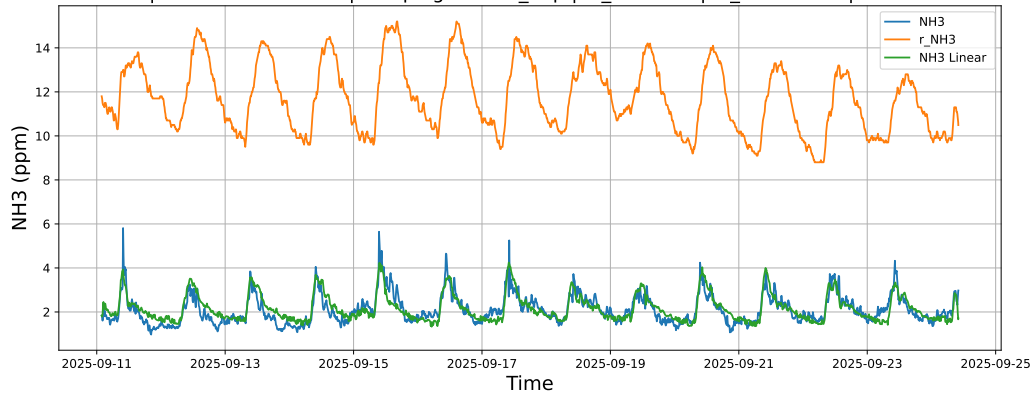


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

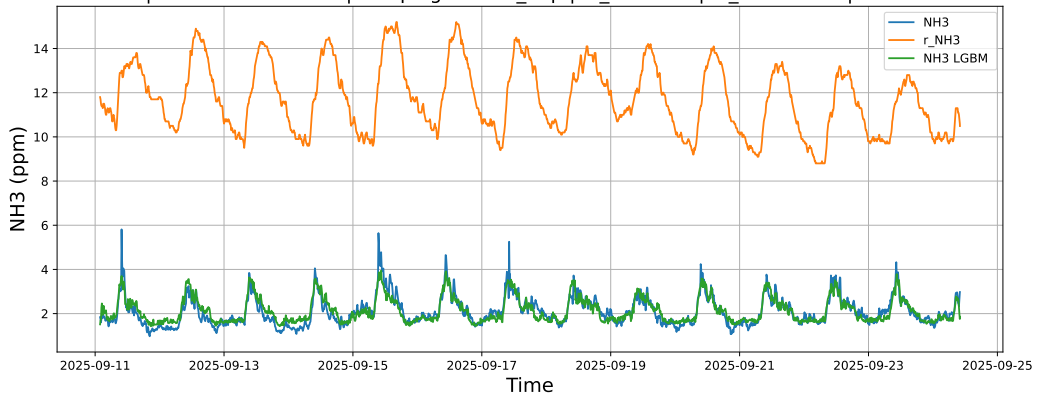


4.1.7.3 Time window : ALL

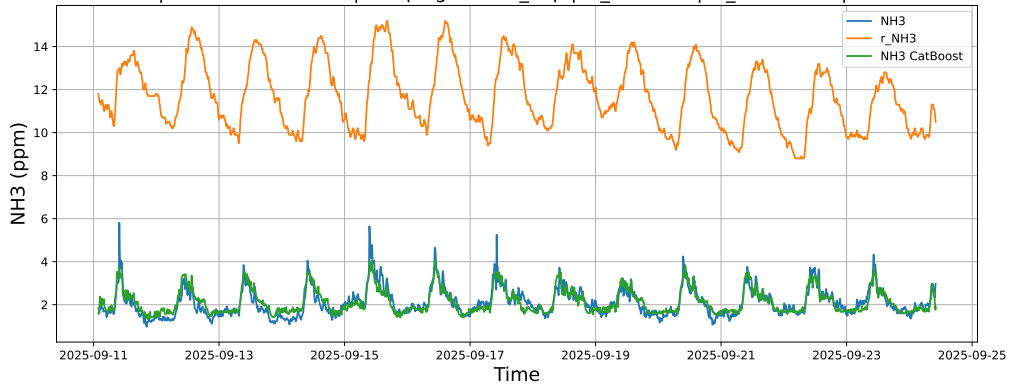
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



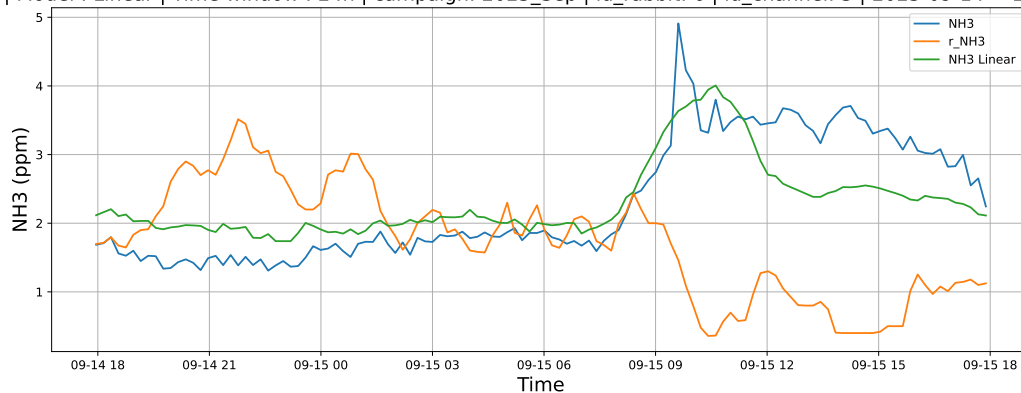
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



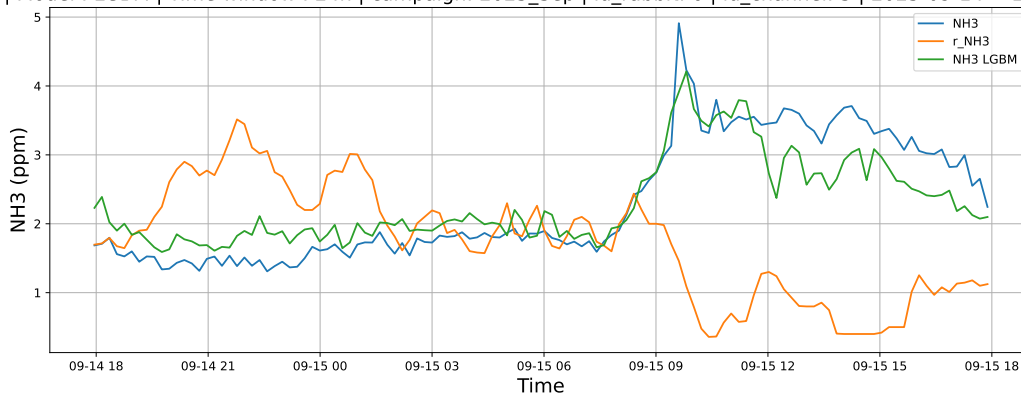
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

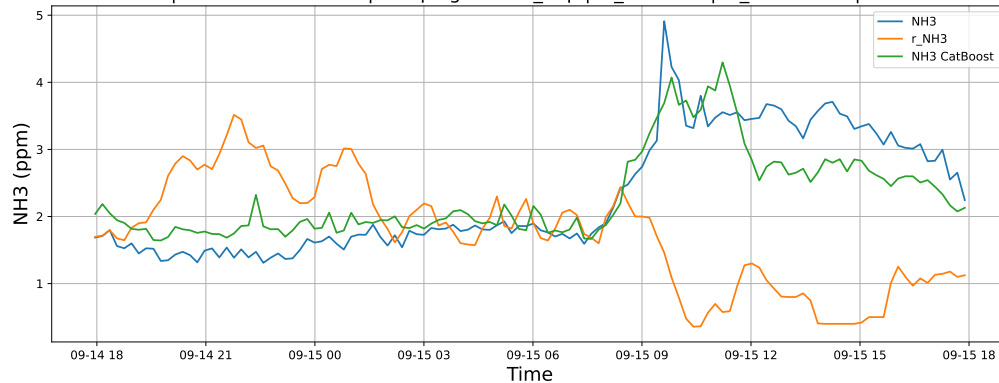
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

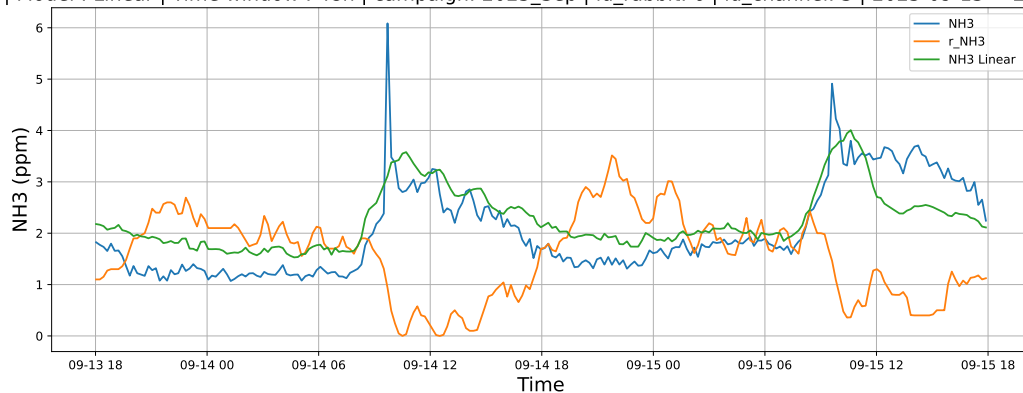


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

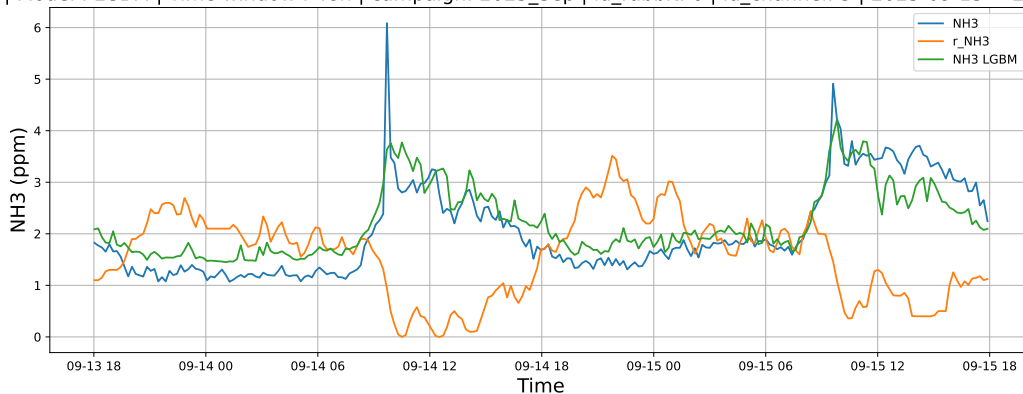


4.1.8.2 Time window : 48

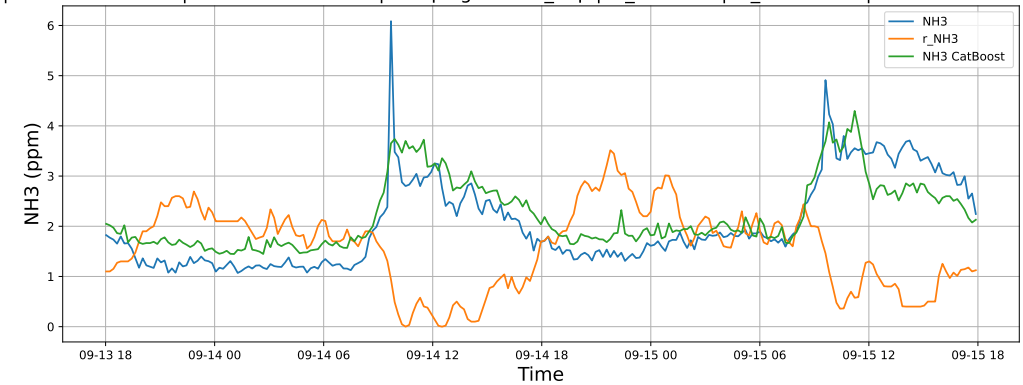
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

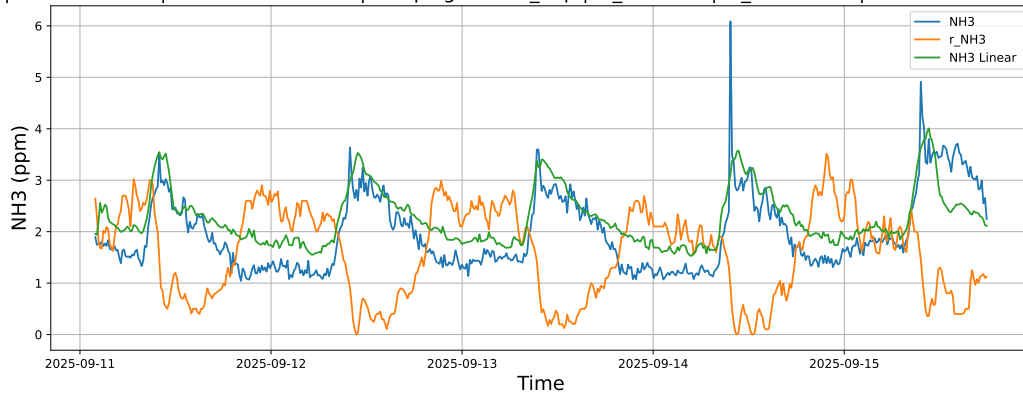


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

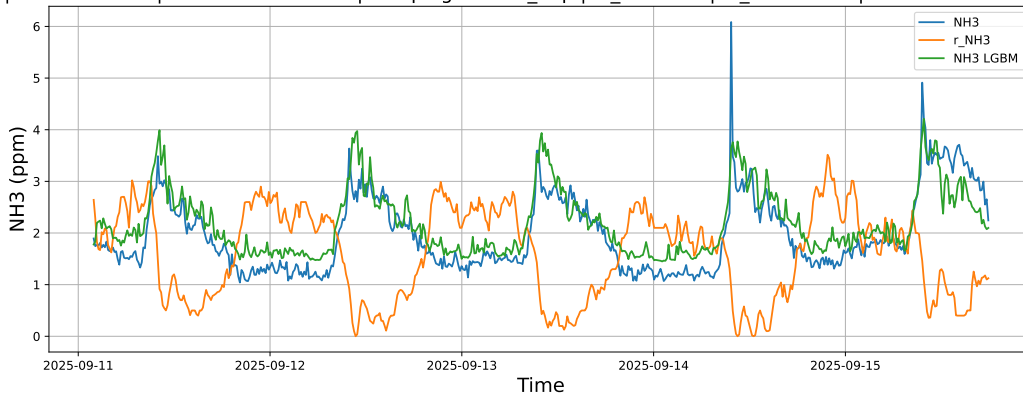


4.1.8.3 Time window : ALL

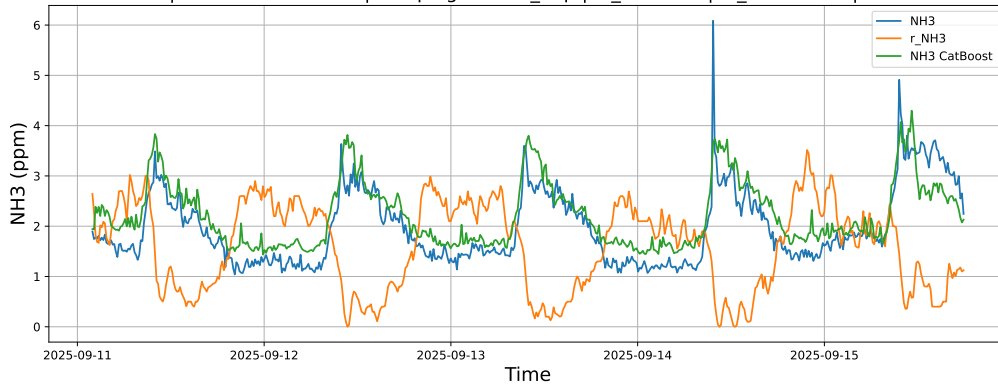
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

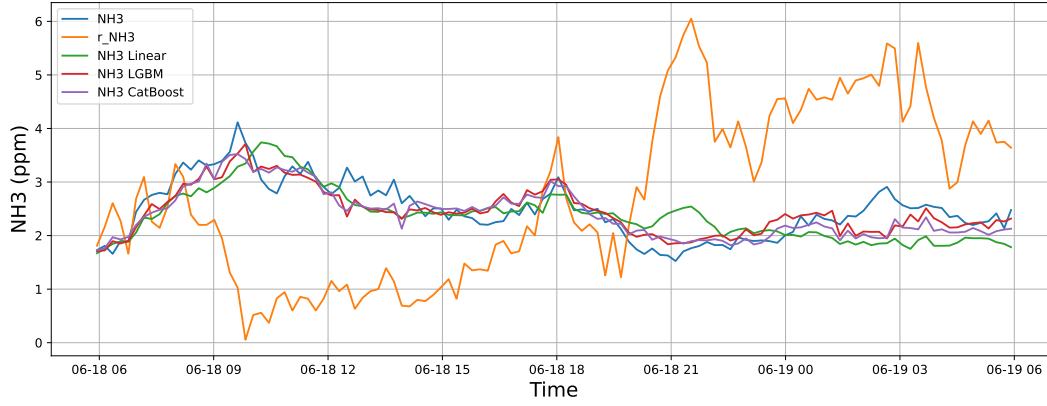


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

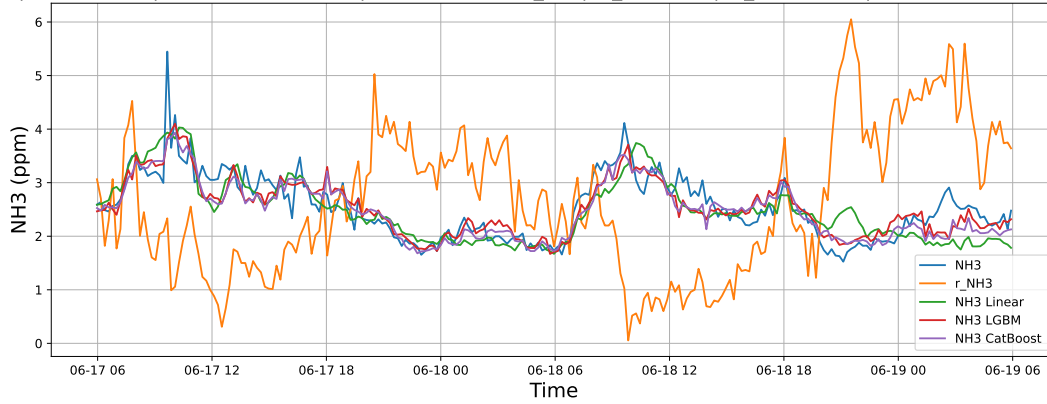
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



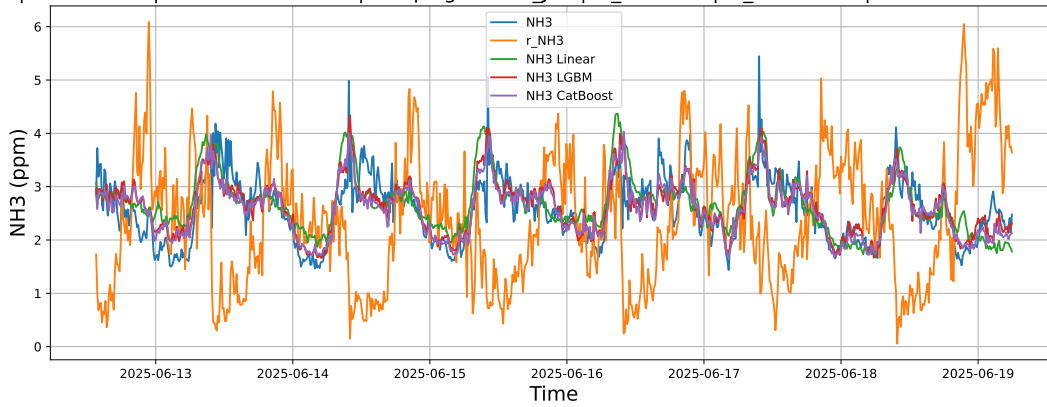
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

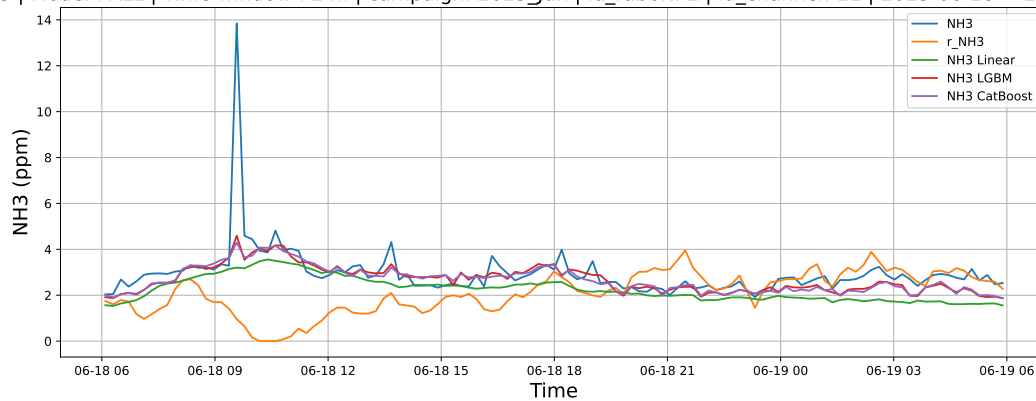
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

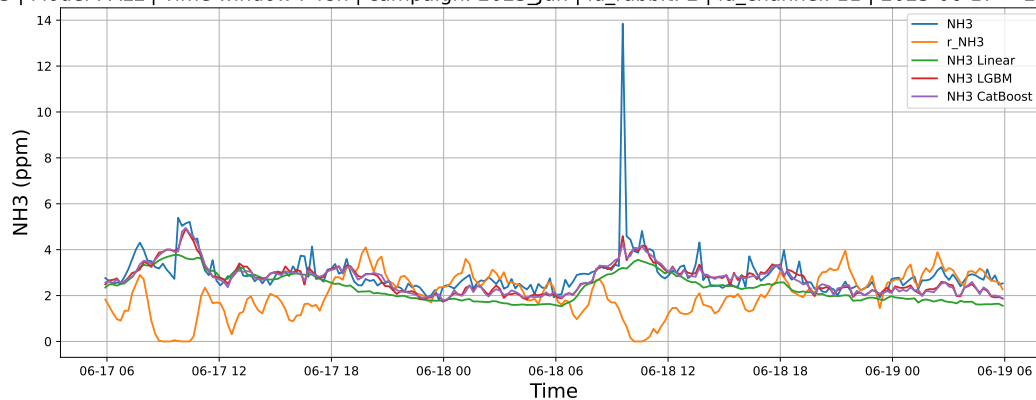
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



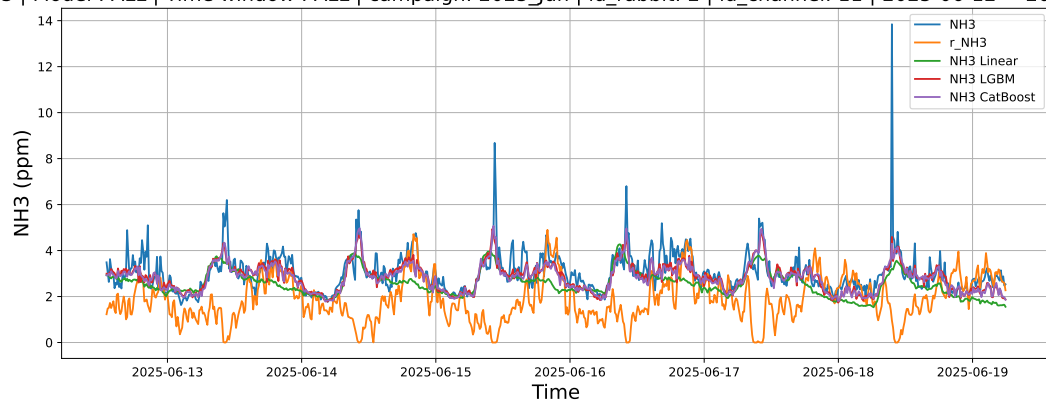
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

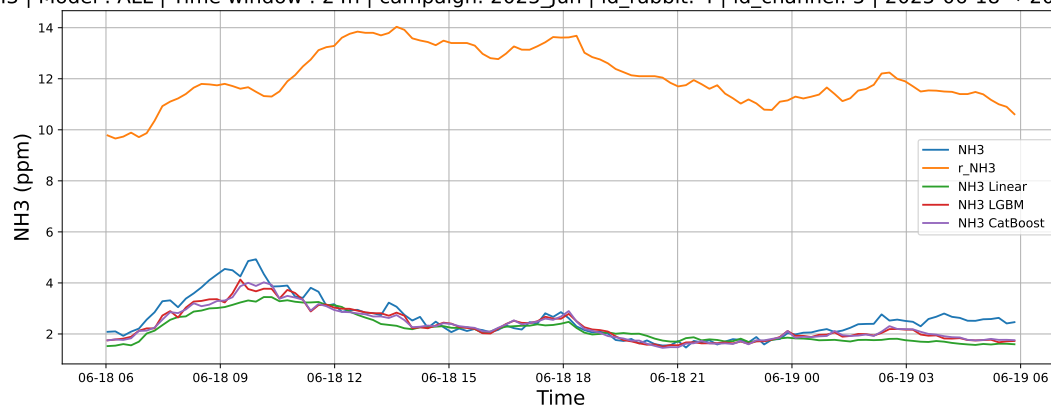
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

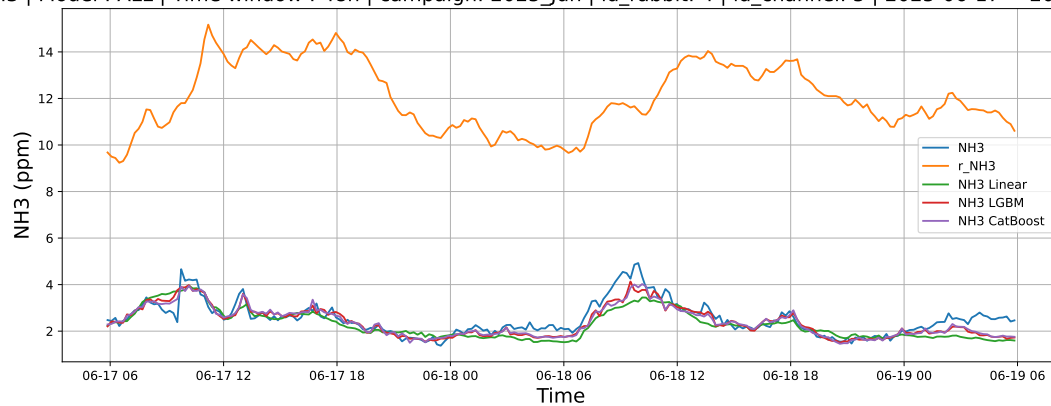
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



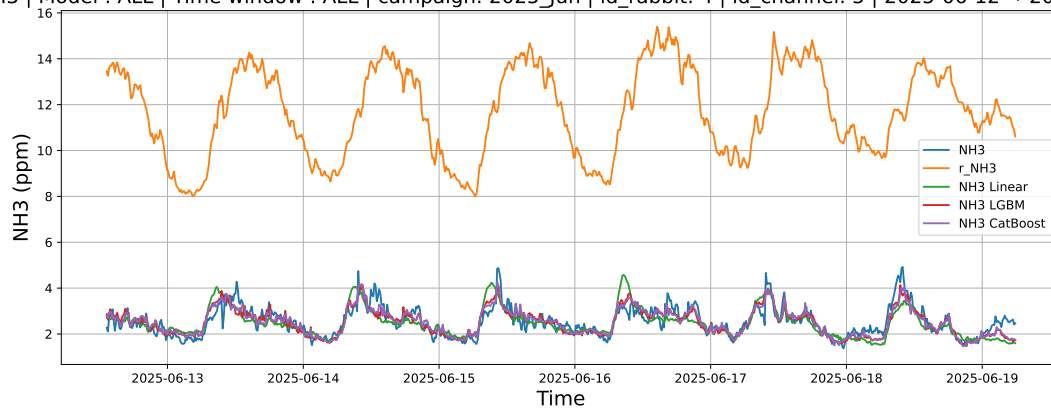
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

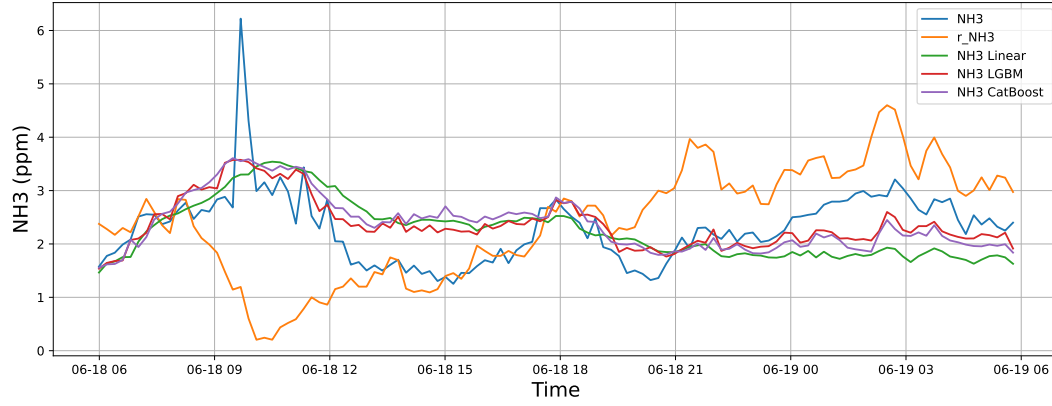
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

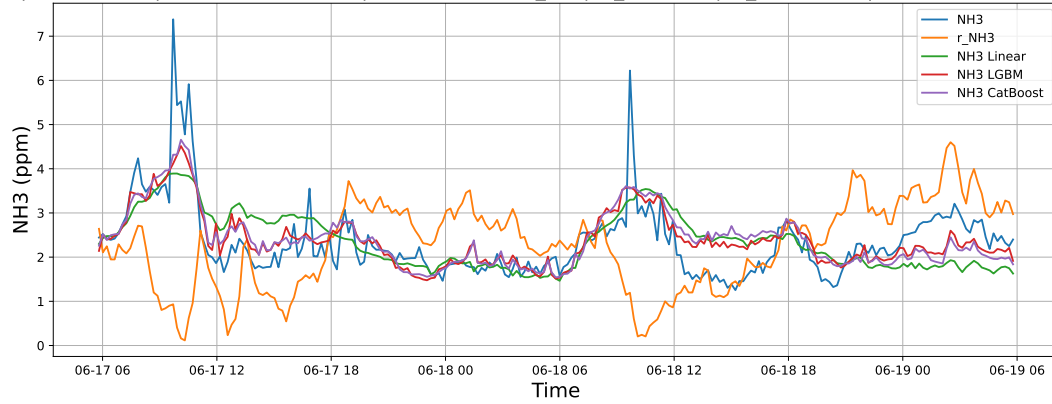
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



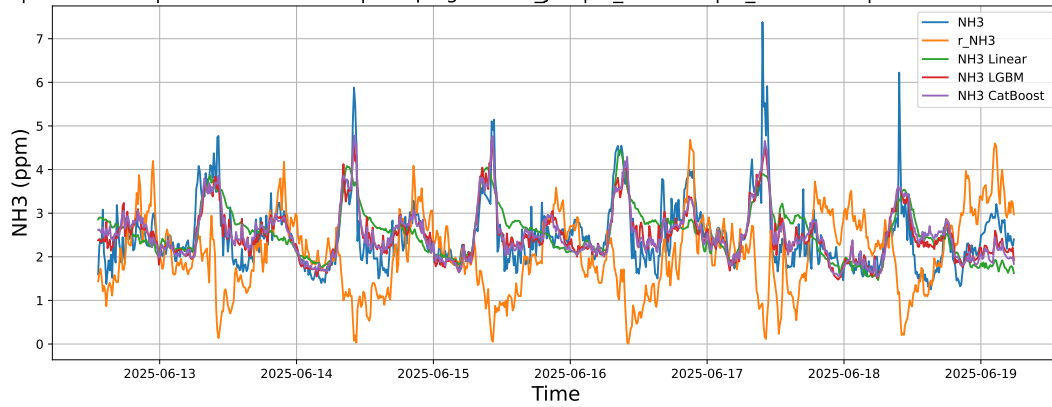
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

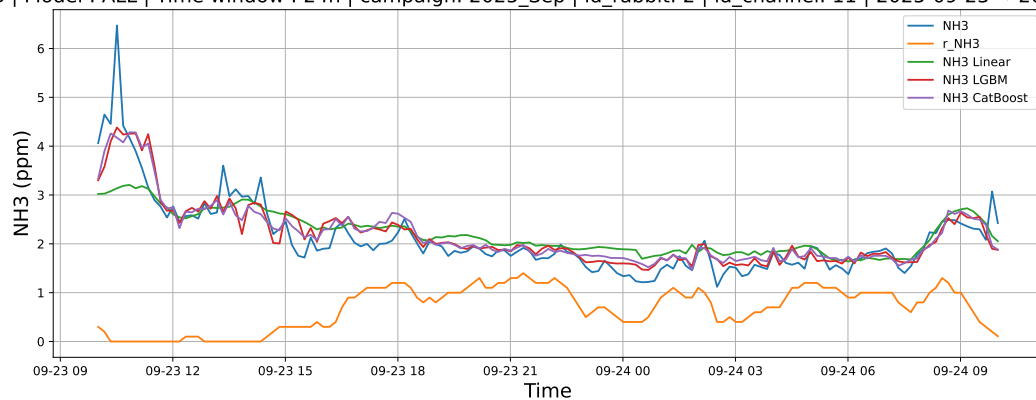
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

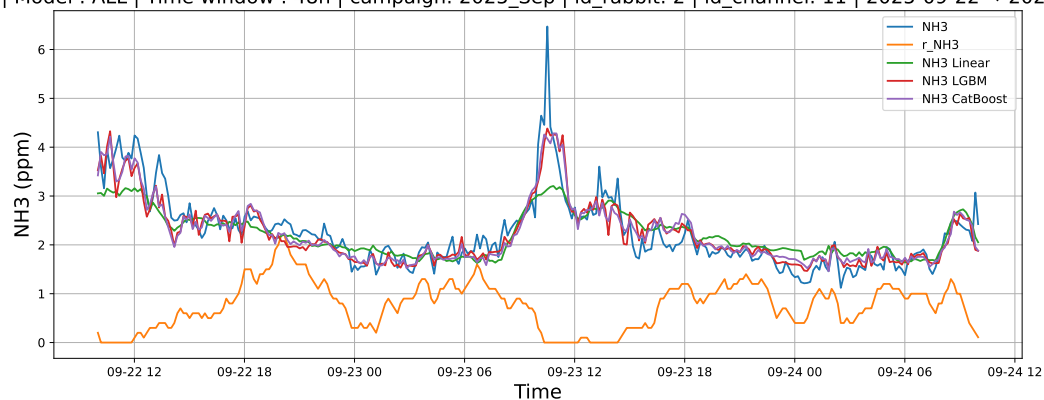
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



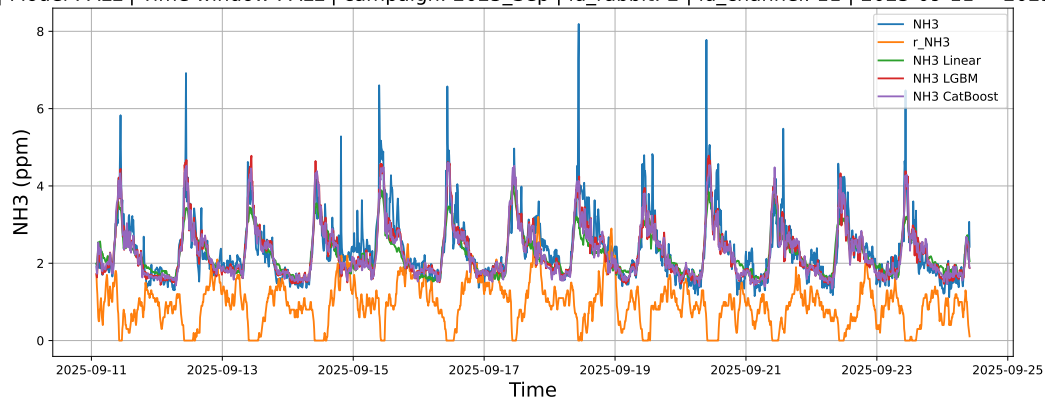
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

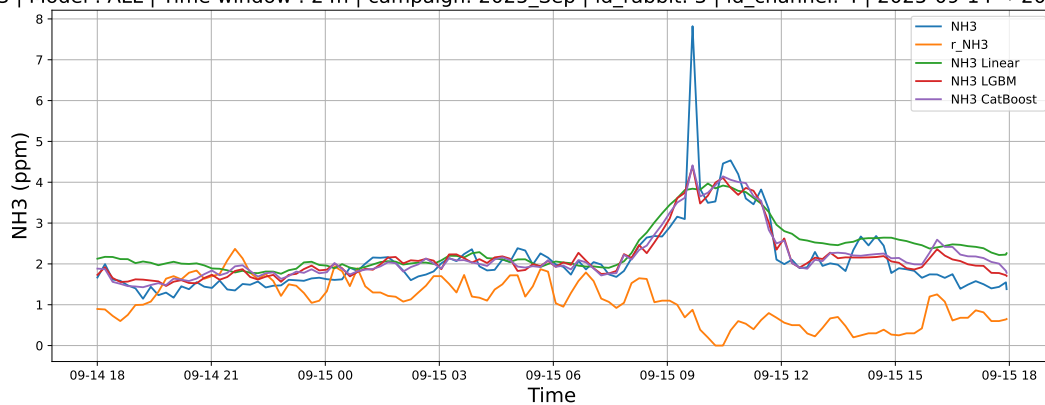
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

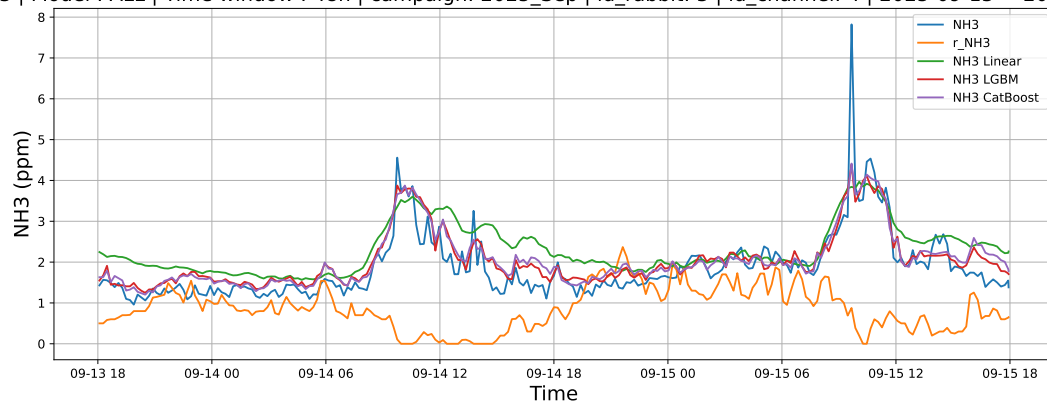
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



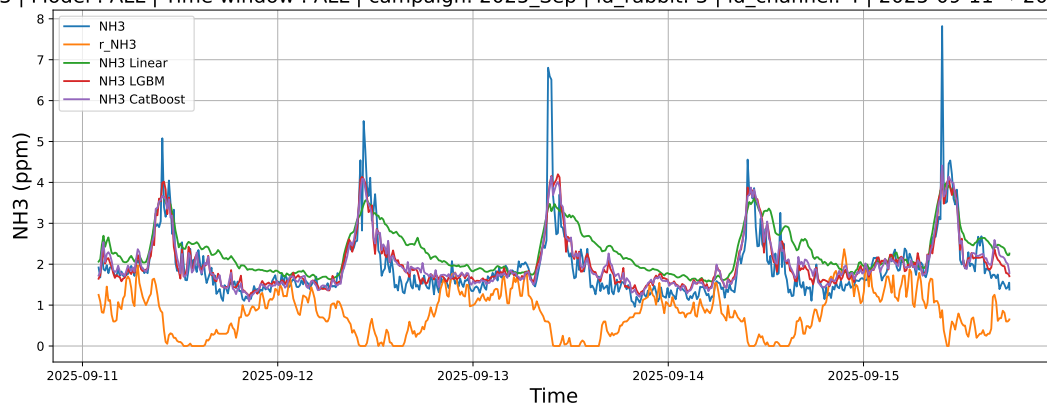
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

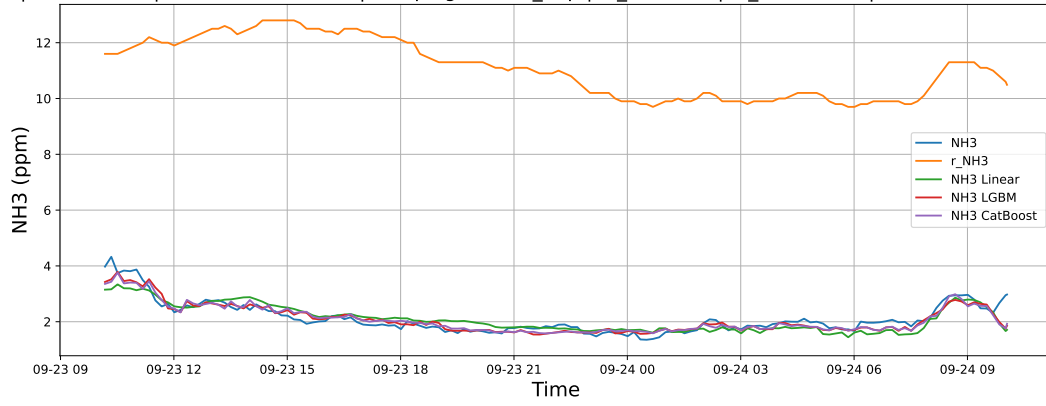
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

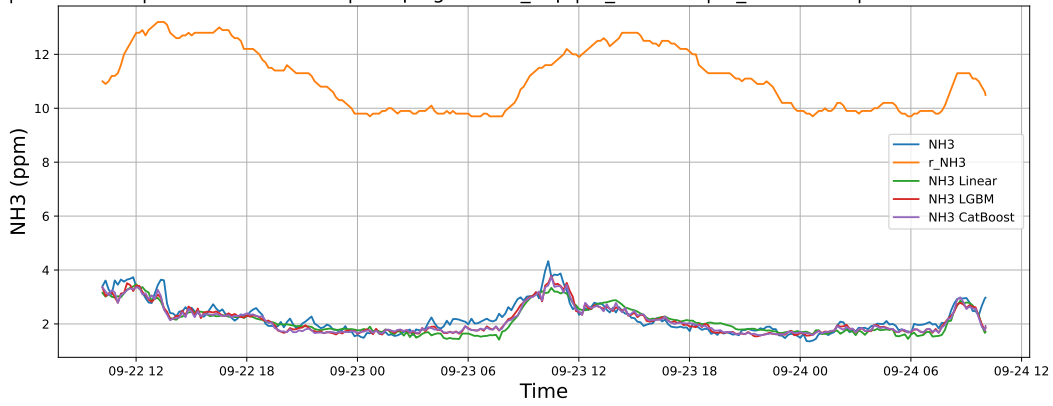
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



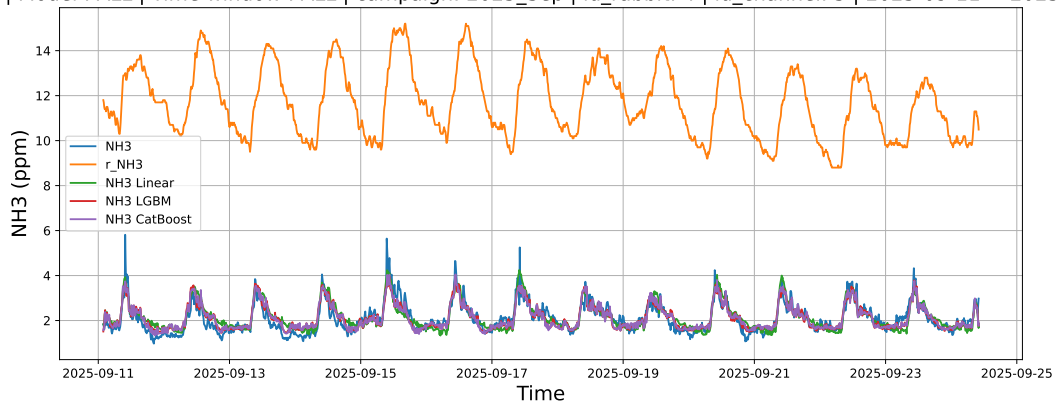
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

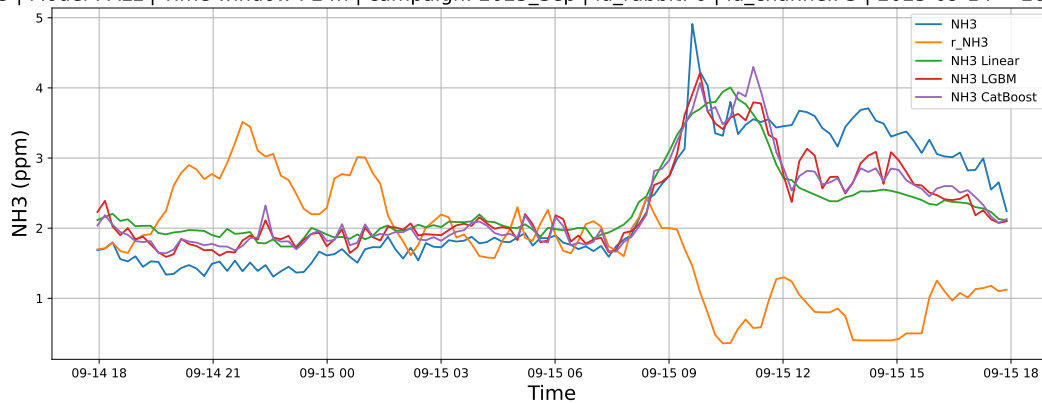
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

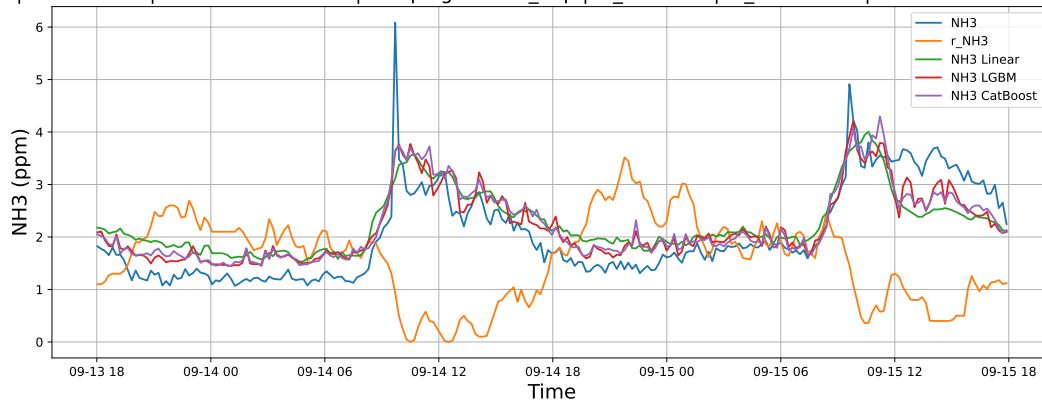
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

